Design Science

Supporting design for additive manufacturing: insights from product development practices in the aerospace industry

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Abstract

Additive manufacturing (AM) enables the production of innovative, lightweight component designs in the aerospace industry. However, AM processes introduce new production feasibility considerations that must be addressed during product development. Therefore, engineers require effective design support and a new design approach to fully exploit AM's capabilities while balancing its constraints. Through an interview study involving 20 AM aerospace industry professionals from nine countries and 10 organisations, this research identifies AM design opportunities and challenges and explores the design supports used to achieve and overcome them. The findings indicate that Laser Powder Bed Fusion is a predominant AM process in aeronautical and space applications. Further, the study identifies practical and computational design supports, describes how AM design is approached during product development and provides a model outlining a general AM design approach. Key AM design challenges identified include insufficient knowledge of material properties, limited sharing of design knowledge and a lack of understanding of the relationship between AM design and post-processing requirements. Consequently, skills gaps and educational needs for Design for AM in aerospace engineering are highlighted. Additionally, the study suggests that further AM aerospace standards, enhanced computer-aided engineering software for AM and artificial intelligence integration could improve design support.

Keywords: Additive Manufacturing (AM), Design for Additive Manufacturing (DfAM), Design Support, Aerospace Product Development, Engineering Design Practice

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1. Introduction

Additive manufacturing (AM), also known as 3D printing, is the process of joining materials layer-by-layer to create parts (ISO/ASTM 2021). Implementing AM transforms product development by creating a more integrated relationship between design, prototyping and production, providing design solutions beyond traditional subtractive manufacturing capabilities. However, AM also introduces new design challenges. The layer-by-layer process imposes constraints, and each AM machine has its own limitations. Hence, engineers require new design knowledge and support tailored to AM's limitations to fully obtain its benefits. While

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various design supports exist, their effectiveness varies and specialised software support remains limited (Wiberg, Persson & Ölvander 2019).

The aerospace sector has been at the forefront of AM adoption due to its demand for high-performance, lightweight components and the economic feasibility of low production volumes. Aerospace refers to the engineering of components used in aeronautical applications within Earth's atmosphere and astronautical applications in space. The industrialisation of AM is predicted to have significant growth potential for aerospace manufacturing (Blakey-Milner *et al.* 2021; London Economics 2019). Metal AM, in particular, offers key advantages for the aerospace industry, including the ability to accommodate complex geometries, reduce weight and enable direct part fabrication, rapid tooling, prototyping and repair (Najmon, Raeisi & Tovar 2019; Gibson *et al.* 2020). With industry growth, the global aerospace parts manufacturing market is projected to increase from approximately USD 900 billion in 2023 to USD 1,230 billion by 2030 (SkyQuest 2024).

Despite this potential, aerospace adoption of AM faces significant challenges. Component designs often require multiple integrated parts that must achieve high performance while balancing lightweight design, cost management and stringent quality requirements (Dordlofva & Törlind 2018). AM process characteristics make achieving this design–quality balance difficult (Dordlofva & Törlind 2018). Key design and process challenges include variability in surface quality, porosity, residual stresses and distortion caused by thermal gradients, all of which hinder performance and increase cost (Calignano & Minetola 2019; Wiberg *et al.* 2019; Artzt *et al.* 2020; Zink *et al.* 2020). Furthermore, these issues complicate the qualification and certification of aerospace parts, requiring engineers to adapt traditional design and product development processes, as design choices in AM are closely intertwined with manufacturing constraints (Gradl *et al.* 2022). Hence, there is a growing need for effective, aerospace-tailored Design for AM (DfAM) support to help designers navigate these constraints and for a clear understanding of how AM implementation influences the overall design approach during product development.

This paper investigates how aerospace engineers adopt and apply DfAM during product development. Drawing on interviews with aerospace AM professionals, it identifies opportunities and challenges, examines current design support and existing gaps and synthesises these insights into an AM design approach model based on aerospace practice. To frame the paper and its contributions, it is first important to outline the current role of AM in aerospace and the process characteristics that shape both its opportunities and challenges.

2. Background

As AM processes have advanced, their capability to produce high-performance, end-use products in the aerospace industry has been demonstrated (Gradl *et al.* 2018; Yusuf, Cutler & Gao 2019; Blakey-Milner *et al.* 2021; Hurtado-Pérez *et al.* 2023). One AM process popular for manufacturing lightweight complex aerospace designs is Laser Powder Bed Fusion (LPBF) (Atkins *et al.* 2018; Najmon *et al.* 2019; Yusuf *et al.* 2019). LPBF uses a laser to selectively melt metal powder layers, forming parts layer by layer (Gibson *et al.* 2020). AM processes, such as LPBF, enable advanced design approaches, including topology optimisation

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(Zhu, Zhang & Xia 2016; Jensen 2018) and part consolidation (Yusuf *et al.* 2019; ESA 2021). Topology optimisation is a mathematical method used in structural analysis to determine the most efficient part geometry (Zhu *et al.* 2016). It uses structural parameters as design variables to improve a part's performance and material efficiency. Part consolidation is a design strategy used in product development to minimise the number of parts. It aims to optimise assembly to improve performance or reduce product costs (Tang & Zhao 2016).

Aerospace companies have successfully leveraged LPBF for these advanced design approaches. Using LPBF, GE produced a fuel nozzle that reduced the number of components from 20 to 1 and weighed 25% less than the previously welded component, reducing their manufacturing cycle time by 40% (GE Aerospace 2018). Similarly, the United Kingdom-based rocket manufacturing company Orbex uses LPBF to manufacture its Prime launch vehicle's nickel alloy main engines. The engines are manufactured as a single piece to eliminate the weaknesses associated with joining and welding. Orbex states that using AM enables them to produce the engines at a 90% cost reduction and 50% time investment savings (Tubbesing 2019; BBC 2021). Meisel *et al.* (2017) conducted an AM redesign case study of NASA's pencil thruster for spacecraft attitude control. They found that their product, redesigned for LPBF, led to an 86% reduction in processing time, a 31% decrease in development time and a 33% decrease in parts compared to the traditionally manufactured product. Meisel *et al.* (2017) apply these achievements for product development to the DfAM guidance used to support their design activities.

For example, with LPBF, the powdered metal substrate results in as-built parts with a characteristic rough surface, particularly in overhanging design features such as the roofs of circular fluid channels (Zhou *et al.* 2021). This surface roughness can negatively impact the mechanical properties of aerospace components by acting as crack initiation sites and reducing the cyclic fatigue life (Kerstens, Cervone & Gradl 2021). Further, the roughness impacts fluid flow through a channel, requiring engineers to use AM-specific design supports that consider the fabrication quality and the surface condition when calculating part performance (Zhou *et al.* 2021).

Consequently, post-processing is often required to improve surfaces. However, various post-processing methods exist, each differing in effectiveness and cost (Kahlin *et al.* 2020). Hence, achieving surfaces suitable for aerospace applications can be expensive and time-consuming (Kahlin *et al.* 2020). Support structures can be included in the designs; however, these increase build time, material usage costs and require additional post-processing for removal (Zink *et al.* 2020).

A comprehensive understanding of design and manufacturing processes is essential for the successful aerospace AM product development. The close integration of design and production in AM directly influences decisions throughout the product development process (PDP), as design choices are closely linked to factors such as surface roughness, which, in turn, impacts product performance and overall costs. To address challenges such as balancing design capability with part quality and optimising support structure use, aerospace designers require effective DfAM support during product development. Therefore, it is crucial to understand the industrial state of the art in these supports and the design approaches taken. By doing so, aerospace engineers can maximise AM's potential,

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overcome key challenges and effectively meet growing manufacturing demands profitably and sustainably.

2.1. Aerospace AM product development

In this paper, the term "*product*" is used in accordance with aerospace quality management standards (ISO 2015; IAQG 2016), where a product is defined as the result of a process and may include parts, components, or subsystems delivered to a customer. This paper focuses on design and redesign at the component and subsystem levels, where AM is currently most applied in aerospace (e.g., brackets, housings and propulsion elements) (Blakey-Milner *et al.* 2021), rather than on whole-system platforms such as complete aircraft or spacecraft.

Due to the close integration of design and production for AM processes, AM designers and manufacturing engineers must collaborate closely for effective AM product development. A generic PDP looks similar to the one described by Ulrich, Eppinger & Yang (2020), shown in Figure 1. It consists of six main phases, progressing as the product's design specification is increased and the initial concepts are developed until the product can be reliably and repeatedly produced.

The Ulrich *et al.* (2020) PDP model begins with the planning phase, where the product's mission statement is defined. Here, product opportunities are identified from a market and business perspective, initial considerations of the product architecture are made and production constraints are outlined before PDP approval. Concept development follows, generating and evaluating product concepts for feasibility and cost. In the system-level design phase, the product architecture is defined, broken into subsystems and suppliers are selected. Next, the detailed design phase finalises part geometry, material selection, tolerances, quality assurance and tooling. In the testing and refinement phase, performance, reliability and durability are assessed, regulatory approvals are obtained and design modifications are made. Finally, production ramp-up begins, where the product is manufactured using the intended system, and any remaining production issues are resolved.

Ulrich *et al.* (2020) identify several key characteristics critical for successful product development, including product cost, development time and development cost.

- **Product cost** refers to the total expense of producing each unit, including capital spending on equipment and tooling.
- **Development time** reflects how quickly a product is created, determining how fast a company can respond to competition or technological advancements and achieve a return on investment.
- **Development cost** represents the total expenditure incurred in developing a product, indicating the level of investment required to achieve profitability.

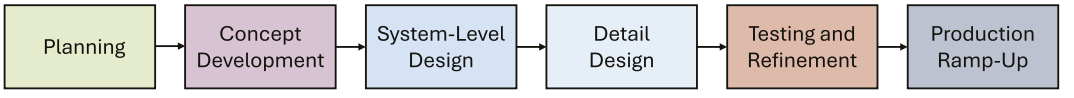


Figure 1. A simplified model of the generic PDP, adapted from Ulrich *et al.* (2020).

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These characteristics serve as essential metrics for evaluating product development performance and are driving factors for AM adoption in aerospace (Blakey-Milner *et al.* 2021). For aerospace, AM offers the potential to reduce product costs by enabling increased design complexity without proportional cost increases (Najmon *et al.* 2019). It also reduces lead times by allowing on-demand part production, eliminating the need for tooling and enabling small and fast production runs (Najmon *et al.* 2019).

Ulrich *et al.* (2020) also emphasise product quality and capability as crucial characteristics of successful product development. AM's alternative design capabilities can be impactful for these characteristics. However, as Ulrich *et al.* (2020) highlighted, product development requires trade-offs. AM engineers face difficulty in understanding design and qualification trade-off challenges like surface roughness (Obilanade, Törlind & Dordlofva 2024). Additional factors such as AM process selection, material properties and product size further impose trade-off considerations (Gibson *et al.* 2020). Optimising for buildability and qualification may come at the expense of part functionality due to limited designer understanding. Hence, engineers need design support to evaluate performance, manufacturability and cost trade-offs.

Ulrich *et al.* (2020) note the process of "*creation*," transforming an initial concept into a physical artefact, as a challenge, as the artefact can often differ significantly from the original idea. AM can enhance creativity by allowing flexible design iterations for innovative solutions. It removes constraints imposed by traditional manufacturing rules and reduces the time and monetary cost of design changes during product development. However, for designers to effectively harness this creative potential, they require support in managing their creative abilities while implementing AM in design (Lindwall 2023). For example, the Manufacturing Process Decision Support (MPDS) developed by Hajali (2024) supports designers in analysing the feasibility and suitability of AM before the conceptual design phase. It advises on the key factors and uncertainties associated with AM, facilitating a better alignment between the concept and the final artefact. This potential for creative freedom can be valuable in aligning the final product more closely with customer needs. While AM may offer creative flexibility more than traditional manufacturing processes, designers must balance creativity with AM's limitations, particularly in aerospace, where the complexity of a part impacts its criticality and must be optimised in the design approach (ECSS 2017). Criticality refers to the severity of the consequences of a potential part failure while also considering the likelihood or probability of its occurrence (ECSS 2017). Additionally, a critical item is defined as one for which it is difficult to demonstrate design performance (ECSS 2023). The more critical a product is, that is the higher the impact or probability of its failure on the mission, the greater the depth and rigour of testing required to ensure its dependability (ECSS 2017). Dependability analysis spans the development process, requiring engineers to integrate fault tolerance and design margins during qualification (ECSS 2017).

AM introduces uncertainty in aerospace qualification (Dordlofva & Törlind 2017; Yusuf *et al.* 2019; Blakey-Milner *et al.* 2021). In the space industry, quality is defined as the degree to which a set of characteristics fulfils requirements (ECSS 2023). Qualification is the first step in verification towards the validation of a part, ensuring that it can accomplish its intended use in the operational environment (ECSS 2023). Qualification is conducted through testing within qualification

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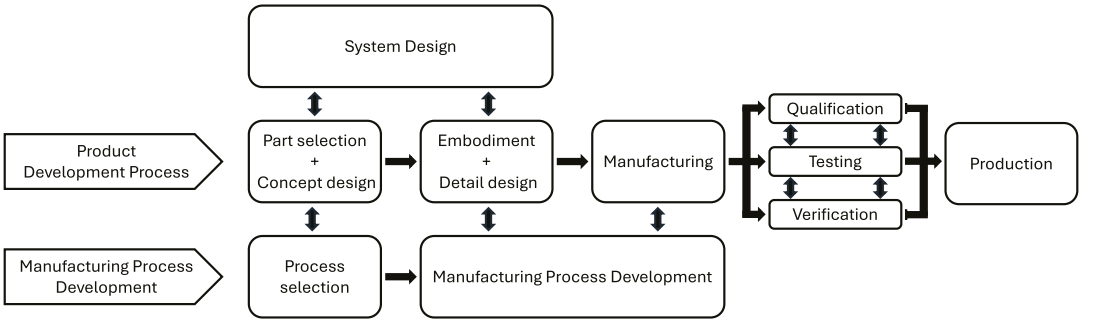


Figure 2. A model of the PDP for AM in space applications, utilising concurrent product and manufacturing process development and systems design. Adapted from Dordlofva & Törlind (2018).

margins, subjecting the product to conditions beyond its operational boundaries to demonstrate the robustness of the design. Several AM process and design factors affect a component's operational performance, which in turn may impact how the qualification of an AM component is secured. Research has explored methods to overcome these challenges by following design supports that guide AM design for qualification (Dordlofva 2020; Borgue, Panarotto & Isaksson 2022) and inspectability (Mahan *et al.* 2022; Cantero-Chinchilla *et al.* 2024).

Borgue *et al.* (2022) propose a model-based design for a qualification method to quantify the risk of AM process uncertainties and geometric variations, enabling more efficient aerospace product concept evaluation by reducing time and cost associated with testing. Dordlofva & Törlind (2017) conducted a study on space industry product development, finding that conventional product and manufacturing process qualifications are linked, with the qualification being to show that the product design intent has been fulfilled. They highlight similarities between AM and welding and casting, where qualification is required machine-by-machine. In consideration of this, Dordlofva (2018) advocates for the concurrent development of AM parts and processes. Thus, AM PDPs must evolve from traditional models to maximise AM's potential. Figure 2 presents a refined model of the PDP for AM space applications, adapted from Dordlofva & Törlind (2018). This model considers concurrent development from two perspectives: designing the part for the process (Seifi *et al.* 2017) and designing the process for the part (Orme *et al.* 2017).

A highly collaborative approach to AM design is needed due to the close relationship between design complexity and buildability. The AM PDP is an inherently iterative process, with manufacturing engineers gaining process insights throughout development and feeding this knowledge into product design. Understanding this new design approach, along with the effective design supports, will help aerospace companies address uncertainties, streamline qualification and leverage AM's benefits in reducing lead times, cutting costs and enabling new designs.

2.2. Design support and challenges in AM

To effectively implement AM in aerospace, engineers rely on design support methods, tools and guidelines to navigate its unique challenges. These supports

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provide knowledge for avoiding and addressing issues such as surface roughness, anisotropy, support structure requirements and costly post-processing (Wiberg *et al.* 2019; Obilanade, Dordlofva & Törlind 2021; Khorasani *et al.* 2022). Effective design supports are tailored to the unique requirements of AM and help designers balance AM's design complexities with its limitations.

AM design support takes many forms, depending on the type of support required, such as guidelines, methods and software. Guidelines assist designers by recommending design limits for different processes, such as reducing surface roughness by specifying a maximum overhang angle of 45 degrees.

(Kokkonen *et al.* 2016; Diegel, Nordin & Motte 2019; Gibson *et al.* 2020). Adam & Zimmer (2014) developed geometry-oriented design rules for AM. Although the rules are described as process-independent, their applicability is constrained by the boundary conditions in which they were created, that is, they are specific to the material, machines and processes investigated (fused deposition modelling, laser sintering and LPBF).

AM design support methods provide a path to effectively applying techniques for increasing product value, such as part consolidation. For example, Orquéra, Campocasso & Millet (2017) provide a method for defining component boundaries within a DfAM process to conduct a multifunctional optimisation of a part's design, giving designers a better understanding of AM process capabilities for product redesign. Additionally, design support can guide the suitability of different types of components for AM based on characteristics that benefit their AM production or the added value that AM can bring to a product (Campbell, Jee & Kim 2013; Klahn, Leutenecker & Meboldt 2015). In both research and commercial industries, advancements have led to the development of AM design and process simulation software that supports designers by simulating product design and production. These tools assess key factors such as design feasibility, production feasibility, costs, part performance and dimensional accuracy. Bikas, Stavropoulos & Chryssolouris (2016) reviewed AM process simulation research and identified three major categories of AM simulation design supports: analytical, empirical and numerical models.

- Analytical models use mathematical analysis based on physics-based laws to simulate AM processes. They are easily transferable to related AM processes but are limited by the assumptions of the underlying physics.
- Empirical models are derived from experimental data, using observed results to determine coefficients and create a model. They are easier to develop than analytical models but are specific to the process, machine and conditions under which the experiments were conducted.
- Numerical models balance the analytical and empirical, starting with an analytical approach and enhancing it with numerical computation.

Wiberg *et al.* (2019) conducted a comprehensive review of research on design methods and software for DfAM, focusing on the role of design engineers and the tools available to support them. Their study examined design automation for metal AM techniques and reviewed commercially available software for simulating AM processes. They found that no commercially available computer-aided engineering (CAE) software currently allows users to input an initial part design and automatically generate a version suitable for an AM process. Instead, users must manually adapt designs to meet AM-specific constraints, which vary by

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machine and parameters such as maximum wall thickness, overhang angles and thermal distortion. These adjustments are made in general-purpose CAD software, relying on the user's knowledge of AM design rules and the software's capabilities.

Additionally, Wiberg *et al.* (2019) found no dedicated CAE tool to verify AM components' key performance properties, such as structural integrity and fluid dynamics. Hence, engineers rely on commercial finite element analysis (FEA) or computational fluid dynamics (CFD) software for design verification despite their limitations in handling AM-specific data. To address these challenges, Wiberg *et al.* (2019) proposed a structured approach to DfAM that maps existing design supports with recommendations for tools and methods applicable to different stages of the design process.

Similarly, Pradel *et al.* (2018a) conducted a critical analysis of the state of the art in DfAM based on a systematic literature review of 81 articles. They developed a framework summarising DfAM across the entire design process, capturing the knowledge needed to support product designers in developing end-use AM components. In building their framework, Pradel *et al.* (2018a) highlighted several limitations of studies addressing DfAM. They found that much of the design support literature did not clearly distinguish whether the guidance was process-specific or machine-specific. This design support limitation leads to a lack of clarity regarding the contexts in which the support rules and knowledge were applicable.

They also identified that most DfAM support fell into one of two categories: one focusing on leveraging the opportunities of AM and the other on overcoming process limitations. Further, they observed a predominance of prescriptive DfAM studies, which may have limitations regarding their applicability to industrial practice. They advocated for more descriptive DfAM studies that incorporate input from practitioners to investigate the implementation of DfAM in practical settings. This type of activity is significant because one of the key challenges in AM growth is the limited communication and data sharing among practitioners (Yhdego *et al.* 2024). Hajali (2024) highlights the large amounts of valuable information generated during the AM process that can be lost if not properly captured. They advocate for the use of product lifecycle management (PLM) tools to systematically record information on AM processes, helping to refine design practices and facilitate the reuse of data to optimise both products and processes.

Pradel *et al.* (2018a) focused on developing knowledge for the series production of components. However, as aerospace components tend to be low-volume and bespoke, industrial research is required to explore variations in the design approach for this type of component to understand how to balance optimisation efforts with product feasibility.

Given the rapid development of AM processes and continuous innovation in the aerospace industry, research like that of Wiberg *et al.* (2019) and Pradel *et al.* (2018a) is crucial for designers and engineers to obtain an understanding of the state of the art in DfAM. Such research is essential for enabling AM's long-term adoption and development. Both Wiberg *et al.* (2019) and Pradel *et al.* (2018a) structured their design processes based on research publications in DfAM. Expanding on this research by gaining a more holistic understanding of AM design

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approaches and the support tools available in industry would be valuable in assessing the current state of the art in DfAM practice.

The development of AM processes for the aerospace industry offers several advantages in manufacturing complex geometries, lightweighting and part consolidation. However, due to its unique process characteristics, AM also introduces several design challenges that impact part performance and qualification. While design support tools exist to help designers implement AM and manage its limitations, current support remains insufficient or fragmented. Challenges remain in clarifying the process and machine-specific design guidelines, improving AM-specific simulation tools, enhancing data capture and knowledge sharing and balancing design optimisation with manufacturability. Therefore, engineers struggle to integrate the AM-specific design considerations effectively.

With AM processes evolving rapidly, design support must continuously adapt to keep pace. Exploring the benefits and challenges of designing AM aerospace components and how they are addressed in the design process would provide valuable insights into existing gaps and best practices. Addressing these gaps is vital for AM adoption in aerospace, where design optimisation must be balanced with product feasibility.

2.3. Purpose of the study

The purpose of this study is to explore how designers are supported and how they approach AM design when developing AM aerospace components. It examines the factors, benefits and challenges associated with using AM for aerospace applications, as well as the design supports utilised, the features that make these supports effective and existing gaps in support. By doing so, the study aims to identify current design support, how they are integrated into the design process and where further development is needed.

The paper begins with a description of the interview methodology, explaining the respondent profiles and the development of the interview guide. Findings from interviews with 20 AM professionals in the aerospace industry are then presented, providing their perspectives on the opportunities and challenges related to AM in aerospace applications. Next, the paper presents the design supports employed to aid in creating feasible component designs, followed by an analysis of the challenges and limitations of current design supports. Additionally, it summarises respondents' perspectives on the impact of component criticality on design decisions before concluding with an overview of identified gaps and future trends in AM design support. This section is followed by a description of the various AM design approaches used by the respondents during product development, which is synthesised into a model of the AM design approach. The paper concludes with a discussion of the implications of the results on the aerospace AM PDP, followed by conclusions and limitations that outline future work.

3. Method

An explorative interview study was conducted on AM aerospace product design and development to achieve the study's purpose. The study aimed to explore the benefits and challenges of AM in aerospace, the design supports employed, the features that make AM design supports effective and where support is lacking.

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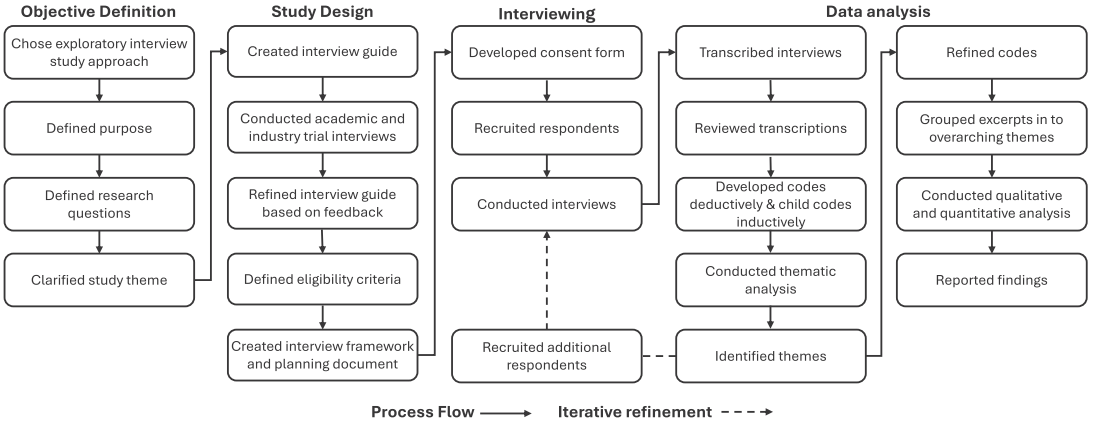


Figure 3. Overview of the explorative interview study process.

The empirical data were collected through interviews with representatives from organisations that produce products using AM. Interviewing was chosen as the data collection method because it facilitates a deep understanding of a subject's viewpoint (Brinkmann & Kvale 2009). In this case, the study examined the state of AM design for aerospace components through the perspectives of aerospace professionals actively designing and producing AM products. This work complements interview studies on DfAM practice, such as Pradel *et al.* (2018b), who investigated professional AM design approaches for series production components. Figure 3 provides an overview of the research method.

The interview guide was created by first thematising to formulate research questions and clarify the theme of the study as suggested by Brinkmann & Kvale (2009), and it followed a methodology similar to Kallio *et al.* (2016). The overall research questions for the study were as follows:

1. What are the current applications of AM in aerospace, and what opportunities and challenges do they present for product development and design?
2. What types of design support are available to industry professionals for AM aerospace components, what determines their effectiveness and what gaps exist in current design supports?
3. How does AM influence the overall design approach during product development?
4. Given the stringent requirements of aerospace components, how does component criticality affect design considerations for AM?

Once the first draft of the interview questions was completed, they were trialled with two academics and an aerospace AM expert to evaluate how respondents could interpret the questions. Based on their feedback, the questions were refined to remove ambiguity before respondent recruitment began. An interview framework and planning document were created to ensure an organised recruitment, interviewing and analysis approach. The document outlined the purpose and interview guide, providing a clear and organised framework to ensure consistency across interviews. Further, having defined the central themes in the document, the framework served as the initial reference for the coding strategy from which the initial codes formed. Therefore, it ensured the consistency of the interview insights and made it easier to compare the data to identify patterns.

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3.1. Sampling and interview procedure

To ensure focused results and an achievable objective for the interview study, eligibility criteria for respondents were established:

1) Respondents must be actively working in the aerospace industry.
2) Respondents must have direct experience in AM product development or process development.
3) Respondents must have worked with AM products produced within the last 5 years.
4) Respondents must have experience designing aerospace components for traditional manufacturing processes.

Due to the rapid rate of development of AM, this sampling criterion would ensure that a state-of-the-art understanding would be obtained from the expert respondents. As it is impossible to sample all professionals who meet these criteria and this is an exploratory study, a non-probabilistic convenience sampling was chosen as the most appropriate way to recruit respondents, like the work of Pradel *et al.* (2018b). Multiple recruitment methods were used to recruit a sufficiently large and diverse group of respondents with relevant experience in AM product development and design:

1) Through attending the International Conference on Advanced Manufacturing 2023 and contacting attendees who gave presentations on LPBF design and aerospace AM product development.
2) Leveraging previous industrial connections at a major aerospace manufacturing company with a large AM department, where the author previously worked. These connections facilitated the identification of respondents at other companies.
3) Through private messages to contacts and LinkedIn members who fit the respondent criteria.
4) A snowballing approach was used, where, at the end of each interview, the respondents were asked if, having now conducted the interview and understanding the purpose of the study, there were any persons they could recommend that would provide useful input.

All respondents were given a consent form to sign outlining the research scope, interview expectations, data management and storage, access permissions and how the interview data would be used. The form ensured that the ethical considerations of the interview were covered, and the respondents were fully aware that the interview was voluntary, that they could withdraw at any time and that no benefit or payment would be received for participating.

The interview questions can be found in the Appendix, and the interview structure was composed of four sections, as shown in Figure 4. The questions aimed to allow for the gathering of relevant information that would enable the purpose of the research to be achieved, that is, to examine how designers are supported in and approach to aerospace AM design.

3.2. Data analysis

The interviews were transcribed and imported into the NVivo 2020 qualitative data management tool for analysis. Thematic analysis was employed to analyse the

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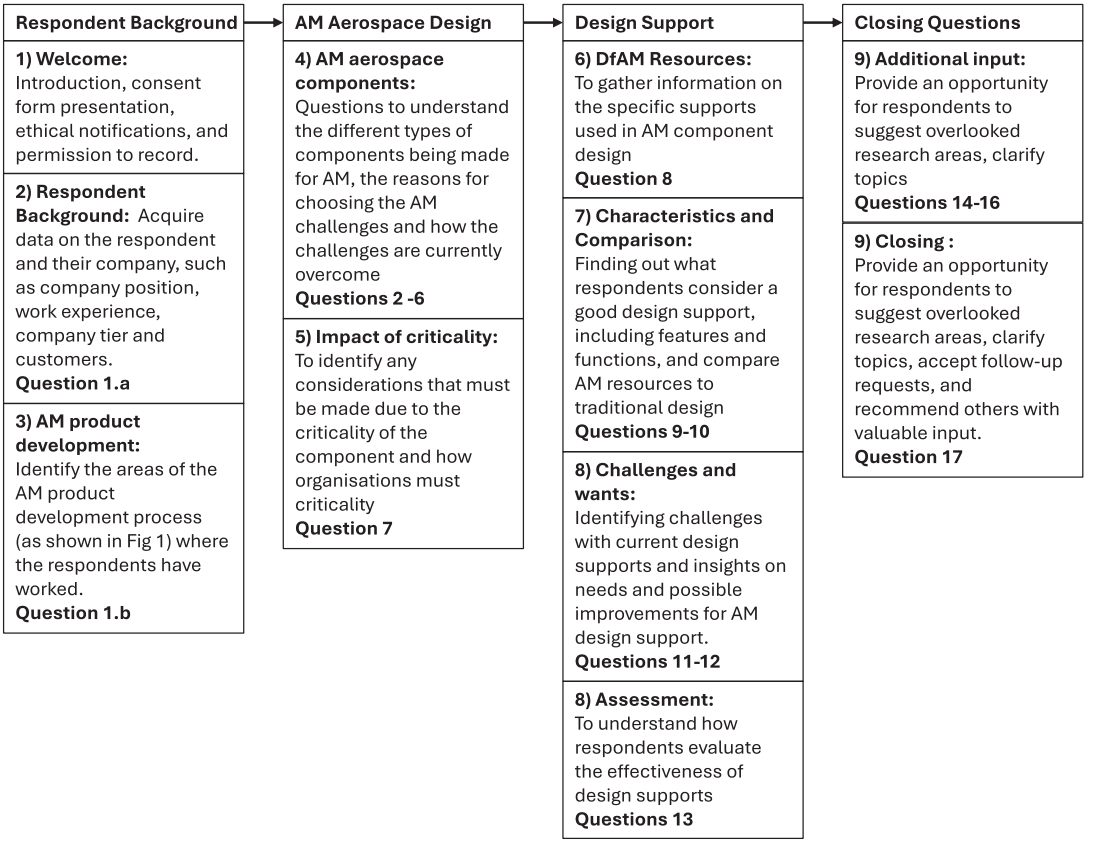


Figure 4. Structure of the interview guide.

transcripts, allowing for the identification of emerging themes and the iterative refinement of codes (Braun & Clarke 2006). Initial codes were reviewed by all authors and developed deductively based on the DfAM and product development literature used to create the guiding document and interview structure. Further intercoder agreement was achieved by having two researchers review the same transcripts and compare coding results, ensuring a shared understanding of the data and the research's reliability. Child codes were developed inductively as they emerged from the data during the coding process. As transcriptions were read and discussed by the authors, new child codes were identified, and a review of previously coded data was conducted to determine if recoding to the child code was necessary. The resulting coding memo, which presents and defines each code, is available in Table A1 in the Appendix. The coded data were reviewed, and thematic analysis led to the identification of five overarching themes regarding AM aerospace applications: Opportunities and challenges, AM design support, gaps and future developments, component criticality and aerospace AM design approach. These themes enable a comprehensive report on the respondents' remarks regarding the current state of AM practice and design support in aerospace product development.

To identify relevant data for each theme, the codes were grouped according to thematic relevance. Each excerpt was then examined for notable details and

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collated into Excel sheets. This process enabled both qualitative and quantitative analysis of the data using pivot tables. Although the data for this research were gathered from individuals with expert knowledge of utilising AM in aerospace applications, it should be acknowledged that the perspectives obtained reflect the respondents' personal experiences and knowledge. As Flick (2009) emphasises, qualitative data, particularly from expert interviews, are inherently subjective and context-specific, focusing not on generalising results but on understanding participants' unique experiences and perspectives. However, the insights derived from these expert perspectives contribute to a broader understanding of AM design challenges and practices, informing a generalised design process applicable beyond individual cases.

4. Findings

Contacting respondents to set up interviews began in February 2024, with the first interview conducted in March 2024 and the final interview conducted in November 2024. The interviews focused on the AM of metal aerospace components, with a further interest in the impact of criticality on a part. While the study initially aimed to cover all metal AM processes, all participants were experts or predominantly experienced with the LPBF process for aerospace components. Some respondents had additional AM experiences in directed energy deposition (DED) and polymer AM. However, most data collected referred to LPBF for metal AM aerospace components.

The respondents are from a diverse range of backgrounds, roles and nationalities. As aerospace is a global industry, getting a range of companies based in different countries and regions was useful. The respondents of the interviews were from a range of organisations, including a small AM consultancy, aerospace manufacturers, an orbital launch services company, governmental and intergovernmental space agencies, defence contractors and an international standards organisation. The country bases of the respondents included India, the UK, Denmark, Switzerland, Sweden, Germany, the USA, the Netherlands and Belgium. As shown in Table 1, a total of 10 organisations were interviewed.

This diversity in backgrounds, expertise and nations strengthens the findings of the industry insights.

The range of experience levels and roles among AM professionals enhances reliability in capturing the state-of-the-art view of AM design in aerospace. The respondents held various roles, from executive and strategic leads such as managing directors and CEOs of AM companies, who can make strategic decisions regarding using different design supports dependent on customer needs and their organisation's objectives. There were research and development engineers who provided valuable insights into the state of the art about design supports. Further, several engineers involved in the day-to-day design and application of AM for aerospace components were interviewed, allowing for a hands-on perspective of the current state of the art.

Much of the data collected by the fifteenth respondent aligned with previous findings, and no additional codes emerged from subsequent interviews. By the twentieth interview, additional data continued to confirm earlier insights, suggesting that data saturation had been reached, indicating rich and reliable findings (Cash *et al.* 2022). Given the rapid evolution of AM technologies,

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Table 1. Table of respondent details for the interview series

Respondent details					Interview details	
ID	Org.	Job title	Affiliation	Aero/space	Venue	Duration (approx. min)
1	A	Co-Founder and Managing Director	AM Consultancy Company	Space	Online	30
2	B	Principal Engineer – Research Specialist in Laser Powder Bed Simulation	Aerospace Manufacturing Company	Aero	Company	45
3	B	AM Research Engineer – Primary Designer	Aerospace Manufacturing Company	Aero	Company	60
4	B	AM Research Engineer – Design Simulation	Aerospace Manufacturing Company	Aero	Company	60
5	B	Senior AM Research Specialist	Aerospace Manufacturing Company	Aero	Company	30
6	C	AM Specialist	Rocket Manufacturing Company	Space	Online	60
7	D	Senior Project Manager	Non-profit Technology Innovation Park	Space	Online	60
8	B	Fatigue and Fracture Mechanics Method Specialist	Aerospace Manufacturing Company	Aero	Online	30
9	E	Design Engineer – Additive Integrated Product Team	Defence Contractor	Aero	Online	45
10	J	Head of Research and Development for Material and Process Development	AM Company	Aero & Space	Online	60
11	F	Principal Engineer	Government Space Agency	Space	Online	60
12	B	Design Leader	Aerospace Manufacturing Company	Space	Online	30
13	G	Director of Technical Operations	International Standards Organisation	Aero & Space	Online	45
14	I	Materials and Process Engineer	Intergovernmental Space Agency	Space	Online	45
15	H	Chief Executive Officer	AM Company	Space	Online	45

Continued

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Table 1. Continued

		Respondent details			Interview details	
ID	Org.	Job title	Affiliation	Aero/space	Venue	Duration (approx. min)
16	K	Lead Mechanical Engineer	Aerospace Manufacturing Company	Space	Online	90
17	L	Customer Success Manager	AM Software Company	Aero & Space	Online	60
18	I	Materials and Process Engineer	Intergovernmental Space Agency	Space	Online	45
19	M	Rocket Propulsion Engineer	Aerospace Manufacturing Company	Space	Online	45
20	M	Rocket Propulsion Engineer	Aerospace Manufacturing Company	Space	Online	45

software and industry practices, perspectives on these topics could have changed over the nine-month interview period. However, reaching saturation indicated stability amongst respondent's views. Additionally, following the interview guide, reviewing transcripts, sending transcripts to respondents for review and conducting coding concurrently with the later interviews helped maintain consistency throughout data collection. This approach facilitated monitoring for any variations in perspectives throughout data collection. Table 1 presents the details of the respondents.

The interviews were conducted either online using Zoom or at the company site of the respondent. Respondents 3 and 4 were interviewed together due to their closeness on their AM projects. This approach helped facilitate discussion on their joint projects as they discussed and debated the points with each other while answering the interview questions. All other respondents were interviewed individually. The interviews were conducted using a semi-structured approach with an average duration of 50 minutes per interview. The interviews were structured around four key sections:

1. **Respondent Background** – To establish an understanding of the respondent's areas of expertise.
2. **AM Aerospace Design** – To discuss the specifics of designing aerospace AM components.
3. **Design Support** – To explore the design supports used and their effectiveness.
4. **Closing Questions** – To ensure all relevant topics were covered and provide respondents an opportunity to share additional insights, ask questions and recommend others with valuable input for interview.

The self-reporting nature of interviews with industry professionals poses a risk of respondent bias, whereby respondents downplay the challenges they have faced

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or overstate their successes in AM implementation. To mitigate this risk, the respondents were informed that strict anonymity would be maintained, and each was provided with a copy of their anonymised transcript. The study fostered an environment where respondents felt comfortable sharing their experiences by revealing only the minimal details necessary to convey their expertise. Further, the diverse backgrounds of the respondents, the semi-structured interview approach and the variety of question topics enhanced the research and reduced bias (Bergelson, Tracy & Takacs 2022). Additionally, respondents provided publicly available references to their work, such as news articles and research papers. Numerous AM matters were identified, many of which were corroborated by multiple respondents, lending stronger support to the accuracy of their reports.

The semi-structured nature of the interview and guiding questions allowed flexibility, enabling each interview to be adapted to the respondent's specific area of expertise or the AM aerospace product they had worked on. Hence, not all the predetermined questions were asked during every interview, the order of the questioning was not always the same and questions were adapted and guided by the respondents' responses (Robson & McCartan 2015). This semi-structured approach facilitated a deeper exploration of AM and design considerations from their unique perspective.

Across the 20 interviews, totalling approximately 990 minutes, the research produced over 110,000 transcribed words of rich qualitative data. Using the initial structure of the questions from the interview guide as an initial framework and the initial data analysis, four distinct themes were developed: (1) Opportunities and challenges, (2) Aerospace AM design support, (3) gaps and future developments and (4) component criticality. These themes provided a structure to the insights of key topics discussed by respondents. Additionally, the design approach taken in product development was highlighted as significantly different from standard practices. Respondents emphasised the need for a more integrated approach to design and manufacturing within product development. This aspect was captured in an additional theme, (5) aerospace AM design approach. Therefore, the findings are provided in two parts:

- The first part is descriptive, capturing and analysing the current state of AM and design support based on the four themes.
- The second part is prescriptive, describing an altered approach to product development shaped by the use of AM, in which design, manufacturing and prototyping are closely integrated

These themes form the framework for presenting the results of this paper, providing insight into the current state of practice in supporting and approaching DfAM in aerospace applications.

4.1. Opportunities and challenges

The first task of the interviews was to gather data on the different products that the industry is focusing on manufacturing to understand the AM applications in aerospace and the factors considered in the design of AM aerospace components. The diversity of respondents from both the aeronautical and space industry manufacturing allowed for a breadth of data collection on a large range of products, highlighting the significant use of and benefits of AM in the aerospace industry for

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several product types. Product types in both aeronautics and space ranged from engine components, structural parts and satellite parts. Half the respondents discussed using AM for a wide range of engine components such as combustion chambers, injectors, nozzles, thrust chambers and turbo pumps (R1, 3, 4, 5, 6, 9, 10, 11, 15, 20). AM was also utilised for structural components such as frames and casings (R1, 8, 14), with a particular focus on using AM for structural brackets (R7, 10, 14, 15, 17, 18). AM was also found to be used for more innovative and specialised applications, such as heat exchangers (R10, 15) and hydraulic components (R16, 20), and within the space industry for specific satellite components such as RF antennae (R10, 14), optical parts (e.g., bi-folds and base plates) (R14) and hydraulic and electrohydraulic actuators for rocket engines (R16, 20).

4.1.1. AM opportunities

The respondents stated several reasons why AM was selected to manufacture these types of products. The main reasons highlighted were the freedom of design and geometry that AM provides over traditional manufacturing, enabling the creation of complex geometries that would not be feasible through traditional means, such as integrated cooling channels within thrust chambers (R6, 20) and topology-optimised brackets (R7, 14). The respondents commented on how the design complexity enabled by AM allows for component designs that focus on weight reduction (R3, 4, 9, 10, 13, 15, 19) and cost reduction (R3, 5, 15, 16). Emphasised by Respondent 6 that "…the business case of AM is the weight savings and the cost savings." Further, Respondent 14 stated that for aerospace AM components, "… the most important points for now [are] lead time and cost."

AM was particularly noted for the weight-saving benefits provided to space industry products (R10, 13, 14, 15, 16, 19). Respondent 10 provided an example where using AM in the production of an antenna cluster resulted in a 10% weight reduction, which they described as having a "huge impact." They also noted the additional functionality enabled by AM, stating, "In a cluster that's printed, you can have more horns in the same space as a cluster that's made traditionally. So, more bang for your buck." Further, they highlighted the advantages of AM part consolidation in transforming a complex assembly antenna design into one part. Hence, it creates a "much simpler system to manage, [with] less chance of failures, mistakes, or issues." However, a balance needs to be considered when leveraging AM's design capabilities for such products. As Respondent 14 explained, the benefits come at a cost, stating, "It's probably a little bit more expensive, but the functionality and the weight reduction helps pay for this."

Respondents noted how AM helps address product development challenges such as time-to-market, long lead times and supply chain issues in the aerospace industry (R1, 2, 6, 8, 15, 16). Respondent 6 compared AM to computer numerical control (CNC) manufacturing, describing AM as "incredibly fast" in comparison. They explained that CNC requires the development of numeric control software, whereas AM allows one to "go straight to production." Respondents highlighted AM's capability for part consolidation methods, emphasising its potential to streamline design and manufacturing processes (R6, 7, 10, 14, 16, 20). Respondent 16 explained that the freedom of shape offered by AM significantly reduces production time. It also enhances the functionality of thrust chambers by enabling the integration of cooling channels into the part. Respondent 16 stated that

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manufacturing a thruster previously required 3–6 months. However, with the design freedom enabled by AM, it is now possible to produce an engine more than 700 millimetres high in just 6 days. Hence, after post-processing, "in less than a month, you have a thrust chamber." Respondent 9 echoed this statement, explaining that AM enables greater "agility" in product development, allowing for rapid iterations of designs and tests due to not having to wait for long lead-time items that require tooling. Respondent 7 provided an example of integrating electrical wiring into a satellite bracket, enabling electrical connectivity without requiring assembly. Similarly, cooling and flow channel integration was a key design freedom benefit identified for AM aerospace applications (R6, 9, 16, 20). Respondent 16 succinctly described the design freedom benefit as "it allows you to, instead of trying to fight against the problem, you can avoid it." They provided an example of hydraulic manifold components, which require multiple cavities for fluid flow. Traditionally, these cavities are drilled or milled, leaving open holes that need to be plugged to ensure durability in extreme aerospace environments. Plugging these holes requires precision machining and specialised plugs designed to withstand such conditions. Respondent 16 said that using traditional methods leads to a 10–15% failure rate for the cavities, meaning high rework and scrap rates. With AM, the cavities and channels of these components can be built directly into the part. Eliminating the need for machining and plugs, reducing the assembly complexity, streamlining the manufacturing process and minimising the risk of rework or scrap while improving component performance.

While the respondents gave several examples of the benefits of AM for different aerospace applications, some noted that AM can be seen as a trend worthwhile to explore without a defined cost or performance benefit. Respondent 4 highlighted that for one of their products, the use of AM was "definitely not the business case" as its use was not financially justified. However, they acknowledged a long-term commercial perspective that it is important to be seen by customers as having AM capability, noting that having "additive parts [in] flight is a big hitter." Respondent 2 explained that AM was primarily just exploring alternative manufacturing methods. Similarly, Respondent 8 describes it as "a way of introducing a new technology." Acknowledging that in these cases, "it was not necessarily cost-effective" and was to "see how it performs with components that are relatively low risk." Table 2 summarises the aerospace products where respondents reported applying AM, along with the specific advantages they identified as driving its selection for these applications.

4.1.2. AM challenges

While the respondents discussed several performance, cost and time-saving benefits of adopting AM, they also highlighted several challenges in realising these opportunities. These challenges relate to technical, operational and economic issues, presenting barriers to adopting AM in the industry. The study identified four key themes of challenges: certification and qualification, design and manufacturing integration, material and process inconsistencies, and organisational and industrialisation issues.

4.1.2.1. Qualification and certification. Fifteen of the 20 respondents commented on challenges regarding the qualification of AM components in the aerospace industry. Issues included the need for extensive testing, the lack of standards to

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Table 2. Summary of AM aerospace product types and AM advantages

Product type	Specific products	Respondent	Reported AM advantages
Engine components (R1, 3, 4, 5, 6, 9, 10, 11, 15, 18, 19, 20)	Injectors	6, 10, 11, 15, 19, 20	Reduced lead times
	Thrust chambers	6, 11, 19, 20	Freedom of design, cooling channel integration
	Combustion devices	6, 9, 11, 18, 19	Reduced failure rate
	Turbomachinery	1, 6, 11, 15	Functional integration, improved thermal performance
	Gas turbines	15	Lightweighting, freedom of design and reduced lead times
	Rocket motors	10	
	Frames and casings	8	
Structural components (R1, 3, 4, 7, 8, 10, 14, 15, 16, 17)	Brackets	7, 14, 17	Integration of electrical wiring, reduced assembly complexity
	Aerostructures	3, 4, 17	
	Hydraulic components	16	Reduced assembly complexity, cost reductions and improved fatigue resistance
Specialised components (R7, 10, 14, 15, 18, 20)	Heat exchangers	10, 15	Weight reduction, improved thermal performance
	RF systems	10, 14	Weight reduction, functional integration, topology optimisation
	Satellite parts	10	Weight reduction, functional integration
	Optical parts	14	Weight reduction, functional integration
	Nozzles	20	Complex geometries, reduced manufacturing complexity
	Bolts, fasteners, rotating parts	18	
	Pressurised systems and tanks	18	
	Slip rings	7	

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support these tests and the difficulty in achieving the repeatability and consistency required for aerospace applications. These challenges make verification and certification more complex. Hence, the respondents highlighted the following qualification-related issues:

- A much greater emphasis on non-destructive testing processes increases costs and requires additional design considerations (R7, 13, 14, 15, 16).
- AM products rely heavily on computed tomography (CT) scanning compared to traditionally manufactured products, which is impractical due to high costs (R13, 17) and a lack of CT standards (R14, 16).
- Qualification processes for AM are less established than for traditional methods, with limited data on mechanical properties (R5), a lack of supporting standards (R6, R7) and high sensitivity to process parameter variations that require time and resource-intensive requalification (R7), all of which pose barriers to industrialisation (R4).
- Current regulatory pathways are not designed for AM, resulting in "an insurmountable amount of qualification work needed" that may take years to complete (R15). This challenge is compounded by strict customer quality assurance requirements that constrain design freedom (R16) and by the already high costs of validation and acceptance in aerospace, which increase further with AM (R19).

4.1.2.2. Design and manufacturing integration. Although AM offers significant design freedom, respondents highlighted the difficulty of balancing this freedom with the manufacturing constraints of AM processes (R2, 6, 13, 17). A key issue is the lack of understanding of how design choices impact manufacturing when using AM. Such understanding can be provided with effective design support; however, the respondents identified a lack of design support that connects specific design parameters, materials and processes. Key challenges regarding design and manufacturing integration include:

- Insufficient support regarding specific machine capabilities (R5, 9, 14).
- Lack of guidance that links design changes to part performance (R6, 10, 11).
- Difficulty in collecting and presenting AM knowledge in a way that is understandable and usable by everyday engineers (R8, 10, 15).
- Rapidly evolving AM processes require continual updates to design support tools (R8), yet these tools often lack reliable material property data (R9) and offer insufficient guidance on how design decisions affect post-processing outcomes (R6).

Respondents 6 and 10 highlighted the challenge of training experienced engineers who have worked with traditional manufacturing processes for years but lack AM-specific knowledge. Additionally, there is a reliance on legacy data from other manufacturing processes when developing AM products (R6, 10). A common challenge identified was the need to design and build support structures to manufacture complex components (R2, 6, 9, 14). Respondent 6 noted that "it's a compromise" when designing support structures, as they impact both product development and manufacturing. Specific challenges regarding support structures include:

- Determining the optimal placement of support structures, especially in aerodynamically sensitive parts, to prevent poor surface finishes (R2, 6).

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- Understanding the trade-offs in post-processing, such as the time required to remove support structures (R2, 6).
- Additional build time is required for support structures (R2).
- Increased material costs due to the use of support structures (R2).
- Lack of design guidelines to aid in support structure optimisation (R6).

CAE supports, such as CFD software, are often used to help qualify components. If the simulation provides good results and can be verified against a component reference model, the simulation can be used towards accepting the component (ECSS 2018). However, for AM products, there is a lack of AM material data available for CAE tools, making simulation-based design support too vague. As a result, these tools cannot reliably predict the robustness and quality of verifications. Nevertheless, when combined with other design support tools, they can assist engineers in identifying and mitigating more apparent issues.

The lack of system-level design visibility available to suppliers compounds these challenges. Respondents highlighted that redesign for AM is hindered by limited information about how components integrate within the overall product system, such as an aircraft (R1, R4). Customers often provide part designs as isolated components, with little visibility of their role within larger assemblies. As one respondent explained, when customers provide designs as isolated components, manufacturers "are not aware of the assembly-level challenges of that component," making it difficult to justify certain AM design decisions (R1). Additionally, a lack of understanding of part interfaces can turn AM optimisation into a "guessing game" (R4).

4.1.2.3. Material and process inconsistencies. Eighteen of the 20 respondents (90%) identified challenges related to material and process inconsistencies in the use of AM. Within the excerpts coded under design challenges (n = 159), 61 (38%) specifically addressed material and process inconsistency in aerospace AM. Of those 61 excerpts, 24 (approximately 40%) were directly concerned with surface roughness. Additionally, 12 of the 20 respondents (60%) described various difficulties in overcoming surface roughness challenges. The identified surface roughness-related challenges included:

- Difficulty in achieving specific surface finishes due to the complexity of the AM process (R5).
- Design constraints imposed by surface roughness (R6).
- Difficulty in measuring and characterising surface roughness (R6)
- Variations in surface roughness across a single part (R10, 14).
- Designing a part to enable effective surface roughness post-processing (R10).
- The impact of surface roughness on material properties and part performance (R6, 20).
- Difficulty in treating surface roughness (R16, 17, 18, 19).

Additionally, machine-related issues were identified, including:

- Difficulties in achieving repeatability and consistency across different AM machines (R20).
- Different AM machines can produce varying results even when using the same CAD file, due to differences in machine calibration, build parameters and hardware configurations. As a result, machine-specific modifications are often

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required during pre-processing and simulation to ensure the intended geometry and performance are achieved (R20).

Other material-related challenges regarding manufacturing issues were also identified, such as residual stresses, pitting corrosion susceptibility, fatigue performance, anisotropy and variations in material behaviour compared to traditional methods. These variations make it challenging to maintain material consistency as the microstructure of AM parts varies due to differences in variables such as cooling behaviour during printing (R10, 13). Respondent 16 also highlighted the difficulty in accounting for these variations due to a lack of data on key material properties, such as fatigue resistance, especially across the range of temperatures required for aerospace applications.

4.1.2.4. Organisational and industrialisation. Additional challenges were identified concerning adopting AM within aerospace organisations and the industrialisation of the processes. The primary challenge identified in industrialisation is the lack of experience among aerospace engineers in designing for AM, leading to problems later in product development (R3, 13, 15, 17, 18). Respondent 18 stated that "the biggest challenge is the lack of knowledge within companies," while Respondent 17 described a "massive skills gap" in the aerospace industry for AM. They further noted that in an organisation of 150 people, only two might have the knowledge required to operate an AM machine. Additionally, the lack of knowledge integration between departments was highlighted. Product designers often do not fully understand AM processes, while process specialists may have little experience with the products they are manufacturing (R13). Respondent 17 emphasised the need to involve design engineers in discussions with sales and business development experts to facilitate interdisciplinary knowledge sharing.

There are specific industrialisation challenges associated with scaling up and integrating AM machines within organisations. Respondent 16 highlighted the cost and complexity of AM technology as a significant barrier to implementation, advising that for AM to be effective at an industrial level, it must be vertically integrated within the organisation. Vertical integration entails multiple stages of AM production, such as design, manufacturing and post-processing, being conducted in-house. This integration reduces an organisation's dependence on external suppliers for AM-related activities during product development. In turn, it improves production and development efficiency by minimising the need for external inputs, lowering transaction costs and optimising the flow of materials and information across the development process (Perry 1989). Similarly, Respondent 8 noted that the absence of in-house AM machines hinders industrialisation efforts.

Operational challenges were also noted, including the handling of highly flammable metal powders. Respondent 16 stressed a lack of safety standards for dealing with these materials, requiring organisations to develop their own adaptations of the European Cooperation for Space Standardisation (ECSS) standards (R6). Lastly, respondent 18 highlighted "image factors" as a barrier to AM adoption. They noted that aerospace companies fear reputational damage if a product they specialise in is manufactured using AM and subsequently fails. Such an event could have severe consequences for the company's business continuity. Table 3 summarises the challenges the respondents identified in implementing AM

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Table 3. Summary of identified challenges of AM aerospace application

Challenge theme	Product contexts/ examples	Key challenge	Respondent Ref
Certification and qualification	Safety-critical aerospace structures (e.g., turbine blades, engine manifolds)	Qualification	6, 7, 10, 12, 13, 14, 15, 16, 18, 20
		Testing	3, 8, 11, 13, 14, 15, 16, 19
		Lack of standards	4, 5, 12, 18, 20
		Pre-certified product modification limits	2
		Verification	14
		Certification	15
		Repeatability and consistency	17
Design and manufacturing integration	Complex, optimised parts (e.g., brackets, flow manifolds)	Design Support	7, 9, 10, 12, 14, 16
		Full system design understanding	1, 4, 5
		Legacy design bias	6, 10
		Design and manufacturing balance	13, 19
		Machine variability	2, 17
		Material printability	20
		Support post-processing	8, 11
		Simulation	9, 18
		Support design and location	2, 6
		Design geometry and material properties relation	19
		Build plate location sensitivity	5
		Operator/designer experience	17
		Balancing geometry and material usage	5
		Support structure design and material usage	6
Material and process inconsistencies	High-performance parts exposed to fatigue/ thermal cycles (e.g., combustor parts, nozzles)	Surface roughness	5, 7, 9, 11, 13, 14, 16, 17, 18, 19
		Material properties	2, 9, 10, 11, 12, 14, 18
		Repeatability and consistency	2, 5, 16
		Process parameters	7, 9, 10
		Post-processing	6, 11, 15
		In-situ monitoring	8, 9

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Table 3. Continued

Challenge theme	Product contexts/ examples	Key challenge	Respondent Ref
		Material printability	4
		Support post-processing	2
		Material behaviour	9
		Pre-certified product modification limits	6
		Operator experience	3
		Process monitoring	8
Organisational and industrialisation	Scaling from prototypes to production while maintaining quality and repeatability	Company experience	12, 14, 16
		Industrialisation effort	15, 19
		Machine scale up	10, 17
		Repeatability and consistency	13, 18
		Legacy design bias	6
		Standards	8

in aerospace, consolidating the thematic groupings and showing their relative frequency across the interviews.

This section has presented several opportunities, challenges and other factors that should be considered when designing aerospace components for AM, addressing the first research question. Despite the numerous challenges to implementing AM identified by the respondents, the potential benefits remain significant. Many companies have successfully flown and launched AM-produced components. (Berger 2023; Relativity Space 2023). Respondents attributed their success in AM product development to the design support available to designers and engineers, enabling them to address and overcome the challenges identified and produce air- and space-worthy components. The following section details the various means and methods of support identified by the respondents during the AM design and PDP for aerospace applications.

4.2. Supporting aerospace AM design

The interviews with the respondents highlighted that overcoming challenges in AM design requires a multifaceted approach. The findings from the respondents identified various design support methods utilised throughout the design and PDP for aerospace applications. These methods were categorised into two types: practical and computational design supports.

4.2.1. Practical design support

Regarding practical design supports, respondents emphasised the importance of iterative testing and validation processes, such as using test coupons and prototyping, to identify and resolve design and manufacturing issues. Respondent 14 highlighted their use of prototypes for process and design validation. An initial

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prototype is used to confirm that the process can produce the part as expected, while subsequent prototypes validate the design and are used for testing. This iterative approach helps address design feasibility issues before advancing too far in the PDP. Other practices target specific design challenges, such as distortion. Respondent 10 noted how distortion can lead designers to "over-design" their parts to mitigate it, cautioning that one can "almost kill a project trying to mitigate this distortion." They advocated for employing established aerospace practices, such as stress relief, alongside design adaptation techniques, like oversizing, to counter distortion.

Collaboration with the Original Equipment Manufacturer (OEM), experts such as senior design engineers and machinists, and certifying bodies like the European Union Aviation Safety Agency (EASA) and Nadcap was identified as a strategy for tackling AM-specific problems. Respondent 2 emphasised that close collaboration with certifying organisations helps ensure they are informed about "how the process has been developed and the material properties and the testing [conducted]," simplifying the certification process when the time comes. Respondent 17 observed that organisations like the American Society for Testing and Materials (ASTM) International are stepping in to establish set guidelines, commenting that, previously, "almost every single aerospace company was doing their own thing." While some standards remain underdeveloped, collaboration with agencies, suppliers, customers and research institutions has helped advance AM design and process understanding, thereby addressing challenges in certification and qualification.

Additionally, team members' practical experience and historical knowledge were cited as valuable resources for navigating challenges. Respondents noted the emergence of a new paradigm for AM engineers, who are now required to have a broader range of expertise. As Respondent 13 explained, "The AM engineer now has to be a very well-rounded person in terms of their experience," due to the intricate connections between design, manufacturing and material properties. They added that "unless you've got an idea of how the material is going to be… you're not going to succeed."

The challenge of aerospace companies having enough knowledgeable engineers to overcome issues can be mitigated through cross-disciplinary collaborative workshops and resource pooling between manufacturers, suppliers and OEMs. Such activities contribute to the industrialisation of AM in aerospace by addressing both knowledge gaps and resource limitations. For instance, Respondent 13 highlighted the importance of education, stating, "We work closely with our AM engineers to ensure they have a deep understanding of the materials and processes." Additionally, resource pooling was identified as a means of improving efficiency and reliability. As Respondent 6 described, "We've developed shared testing frameworks with partners to cut costs and improve reliability." These collaborative efforts not only enhance the skill sets of engineers but also foster cost-effective and robust AM adoption strategies. Table 4 summarises the practical design support approaches to overcome some of the identified challenges for AM in aerospace applications.

4.2.2. Computational design supports

In addition to conventional and practical methods, respondents highlighted several computational design supports that aid in AM design and product

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Table 4. Summary of practical design support approaches to overcome AM challenges

Challenge theme	Design support	Design support features	Impact on design/ product development
Design and manufacturing integration	Internal Design Guidelines (R3, 4, 5, 7, 9, 11, 17)	Provides structured recommendations for design and manufacturing integration.	Streamlines workflows and ensures consistency across designs, reducing errors.
Certification and qualification	Iterative Prototyping (R6, 14)	Validates both manufacturing processes and design feasibility.	Reduces design uncertainty, identifies critical design issues early and accelerates development cycles.
Certification and qualification	Collaborating with Standards Agencies (R2, 14, 17)	Develops and adapts AM-specific standards and testing protocols.	Ensures compliance with aerospace safety and reliability requirements.
Material and process consistency	Stress Relief Techniques (R10, 14)	Addresses geometric inaccuracies and mitigates distortions.	Improves structural integrity and dimensional accuracy of AM parts.
Organisational and industrialisation	Cross-Disciplinary Collaboration (R2, 6, 13)	Facilitates communication and resource sharing among engineers, suppliers and regulatory bodies.	Increases knowledge sharing, reduces costs and improves product reliability and manufacturability.
Organisational and industrialisation	Roundtable Discussions (R2)	Facilitates exchange of expertise between engineers, suppliers and regulatory bodies.	Increases understanding of process capabilities and addresses design challenges collaboratively.

development. Budinoff, McMains & Shonkwiler (2024) highlight the role of software-based manufacturability analysis in enhancing AM design outcomes. Their study found that software tools help designers minimise manufacturability issues and boost creativity in design solutions. The respondents widely recognised simulation software as a critical tool for predicting potential issues during the AM process. Issues such as stress concentrations, distortions and manufacturing feasibility. Respondent 5 explained that simulation tools delve deeper into process aspects, such as simulating melt pools, cooling rates and material grain sizes, to determine whether the product will meet the required material properties. However, despite their usefulness, Respondent 4 acknowledged their limitations, stating that these tools "identify problem areas, but not the magnitude,"

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highlighting the need for additional data to support engineers in understanding and resolving design issues. One such support comes from digital material databases, which provide extensive information on materials and processes. As Respondent 11 suggested, such tools can help refine designs by providing a better understanding of material behaviours. Engineers can trace issues, modify build parameters, or determine heat treatments to improve properties through the knowledge in these databases. Respondent 16 stated that the effectiveness of computational design support depends on having "a very good database of material performance" covering variations in manufacturing methods, surface finishing and temperature. However, achieving the required level of detail would, as Respondent 6 remarked, require "either a huge database or an army of PhD students."

CAE tools were identified as crucial for preparing parts designed and optimised for AM production. One example of CAE AM design support is build preparation software like Magics, which was widely used by Respondents 3, 4, 5, 15 and 17. These tools are designed specifically to address AM challenges, offering features such as support structure generation, part nesting and distortion compensation.

Respondent 10 highlighted its role in optimising part orientation: "It ensures we can avoid recoater collisions and minimise distortion while printing." Similarly, Respondent 16 emphasised the broader impact: "Orientation impacts everything: material use, time, and final quality." The effectiveness of these designs supports lies in their ability to address critical AM challenges. The design supports that highlight potential failure areas, such as material property issues, orientation concerns, or distortion predictions, are highly valued, even if their magnitude estimations are imperfect. Furthermore, support with cross-functional integration that seamlessly connects design, simulation and manufacturing workflows is considered essential for improving productivity and reducing errors. A summary of the computational design supports for AM identified from the respondents is presented in Table 5.

4.3. Gaps and future developments

Despite the utility of current tools, respondents highlighted several gaps in design and manufacturing knowledge and design support. These gaps point to areas where future development is needed to enhance the effectiveness of AM for aerospace product development.

4.3.1. Design collaboration community of practice

The primary gap and area for future development identified by respondents is the creation of a design collaboration community (R6, 8, 9, 10, 13, 16, 17, 19). Forty percent of respondents commented on the need to collate and share knowledge through documentation, workshops and collaboration between aerospace organisations and academia. Respondents argued that such a community of practice would help build the skills necessary for engineers and address the lack of operational experience, which often leads to over-engineered parts. As one noted, a catalogue of "intelligent and elegant solutions" would allow others to reference best practices (R10). However, concerns were also raised about the reluctance to share data between organisations (R9, 17) and the difficulty of ensuring data quality (R9). Despite this, respondents highlighted the openness of the additive

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Table 5. Summary of computational design support approaches to overcome AM challenges

Challenge theme	Design support	Design support features	Impact on design/product development
Design and manufacturing integration	Simulation software (R2, 3, 4, 7, 8, 9, 10, 11)	Predicts stress concentrations, distortions and manufacturing feasibility and simulates melt pools, cooling rates and material grain sizes.	Identifies potential design issues early and supports material property evaluation and design feasibility, reducing uncertainties.
Design and manufacturing integration	Material property databases (R6, 11, 16)	Provides extensive data on materials, processes and surface treatments.	Improves the accuracy of simulations by providing reliable material data, aiding in material selection, test and process planning
Design and manufacturing Integration	Topology optimisation (R1, 2, 7, 10, 16, 19)	Enables weight reduction and part consolidation in high-performance designs.	Facilitates the development of lightweight, optimised designs that fully exploit AM capabilities while adhering to functional requirements.
Design and manufacturing integration & material and process consistency	Integrated CAE packages with AM-specific modules (R4, 5, 17, 20)	Combines simulation tools, topology optimisation and material databases	Reduces errors from file transfers, streamlines workflows, improving efficiency and ensuring accuracy during the design process.
Material and process consistency	Build Preparation Software (R2, 4, 5, 6, 8, 10, 13)	Tailored AM features, such as support structure generation and modification and part orientation functions.	Optimises build orientation minimises distortion and recoater collisions to improve final product quality.

community as a strength that could be leveraged to facilitate such collaboration (R13). Respondent 17 noted that a community of practice is beneficial given the high turnover of AM experts and the ongoing need for workforce training.

4.3.2. Integrated CAE and simulation environments

Respondents consistently emphasised deficiencies in current CAE and simulation support. Problems with file conversion, scalability, data loss and inefficiencies when transferring between tools were widely noted by respondent 5, with respondent 4 stating, "The transfer of files is poor. You get a lot of de-featuring, a lot of errors." Additionally, due to the complexity of AM products, CAE software often struggles. Respondent 4 further explained, "Once you get to a certain level, it's just painfully slow."

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Others highlighted traceability challenges, where each hand-off between tools "costs traceability and data quality" (R15). Respondents therefore expressed a strong need for an integrated AM CAE suite that combines simulation tools, databases and topology optimisation within a single package, enabling data integrity and direct export to AM machines (R5, 14, 15, 16, 18, 20). Improving simulation accuracy, particularly for stresses, microstructure, cooling rates and surface roughness, and addressing the "limited interoperability," as highlighted by Respondent 16.

4.3.3. Simulation software and material properties data

Respondent 2 highlighted the need for co-development of AM software between engineers and providers to address aerospace challenges and enable design techniques such as topology optimisation. Respondent 10 bluntly stated that current AM "topology optimisation tools are s**t" for aerospace needs, while Respondent 16 noted that no topology optimisation software has yet been certified for aerospace. Respondent 18 further stressed that not only the AM material and process, but also the engineering method itself (i.e., the topology optimisation process), must be qualified.

Respondent 14 emphasised that improved software, robust simulations and extensive datasets can support quality assurance by enhancing traceability and managing variability between data, models and printed parts. They further underscored the role of digital twins in modelling AM product development and providing quality assurance.

The area's most frequently suggested for development were data and simulation tools to predict and manage surface roughness issues in design and manufacturing (R6, R7, R12, R14, R16). Respondent 6 stated that surface roughness issues are currently time-consuming to overcome and that addressing them effectively during design largely depends on the engineer's experience. Development of surface roughness measurement methods was also proposed (R14, R16). Respondent 14 commented that standard measures such as Ra and Rz are machining-centric and suggested Svk as an alternative, which may better capture fatigue-relevant properties (R14). Better modelling of roughness during design, combined with robust measurement methods, was seen as crucial to improving certification and qualification. Respondent 12 identified this as a key need for AM, given the close link between surface roughness and material properties.

4.3.4. AI generative design tools

Respondents highlighted the potential of developing design databases and artificial intelligence (AI)-driven generative design (R7, 10, 15, 3). Respondent 17 emphasised that AI will be the "key to success" for AM, as it could reduce reliance on specialists and extensive process and design education. Although generative AI tools are emerging, respondents noted several capability gaps. Respondent 19 highlighted the lack of support for transitioning between AM and traditional manufacturing, as well as the limited and immature integration of multiple processes in tools such as Fusion 360. High costs, reliance on proprietary tokens and dependence on cloud computing were also seen as barriers to adoption. Additionally, respondents raised concerns about ownership, expertise and the justification of AI-generated design decisions. Respondent 15 described fragmented workflows as

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the "Achilles' heel" of generative design, noting that every data transition "costs traceability and data quality," undermining certification. At the same time, they acknowledged the promise of these tools, observing that they can "quickly generate" complex geometries, such as lattices, that would otherwise take weeks to model in traditional CAD.

4.3.5. Summary of identified gaps and future developments

The insights provided by respondents suggest several other gaps for future development in AM knowledge and design support, including in situ monitoring (R2, 11), support structure design guides (R12) and the development of additional standards (R8, 11). The identified gaps, support development areas and future trends identified by the respondents are summarised in Table 6.

This section has detailed the various design supports aerospace professionals use to develop feasible and AM-suitable product designs. It highlights how these tools help in understanding AM benefits while accounting for the process's unique constraints. Tables 4 and 5 summarise the features of these design supports, demonstrating how they effectively assist engineers in the design process. Additionally, Table 6 summarises the identified design support gaps, highlighting

Table 6. Summary of identified gaps and future trends

Gaps and future trends	Respondent ref
Design collaboration community of practice	(R6, 8, 9, 10, 13,16,17,19)
AI generative design tools	(R3, 7, 10, 11, 13, 15, 19)
AM single suite CAE	(R5, 14, 15, 16, 18, 20)
Roughness data and simulation	(R6, 7, 12, 14, 16)
Databases	(R1, 11, 14)
In situ monitoring	(R2, 11)
Standards development	(R8, 11)
AM-specific CAE packages	(R1, 10)
CAE cross-compatibility	(R5, 7)
Improved simulation	(R1, 14)
CAE processing requirements lowered	(R15, 20)
Qualification design support	(R9)
Design traceability	(R5)
Software co-development	(R2)
Digital twins	(R14)
Support structure guides	(R12)
Machine calibration methods	(R2)
Classification design guidance	(R8)
Melt pool simulation	(R2)
Mixed material LPBF processes	(R10)

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missing knowledge and areas for future development to create more effective design support. Thus, this section successfully addresses the second research question.

4.4. Component criticality

All 20 respondents used the notion of critical components in some form concerning aerospace products. Criticality was generally regarded as the classification or ranking of components based on their level of importance. Organisations commonly employed formal classification systems, such as class 1, 2 and 3 categories (R2, 3, 4, 11, 13, 14, 17, 18, 20), assessed using a combination of formal standards, customer input and internal processes. Respondents referenced established standards, including ECSS (R6, 14) and ASTM F3572 (R13), as well as aerospace-specific guidelines from organisations such as NASA and EASA (R3, 4, 13, 16). Classification systems such as ECSS (2017) standards are used to rank components based on the consequences of failure and distinguish between mission-critical and non-critical components (R11, 14, 16, 18, 20). For AM service providers and subcontractors, this classification was often tied to customer-driven requirements, where customers specified the criticality of their component (R1, 7, 10, 15). The criticality assessment also considered performance expectations, operational environments and the risk of failure, considering factors such as stresses, vibrations and material properties. Critical components were universally associated with higher risks, including mission failure, safety hazards, or catastrophic outcomes.

When approaching AM design in the aerospace industry, there is a disparity regarding the impact of criticality on the design approach. Sixty percent of respondents indicated that criticality affects their design approach, while the remaining 40% stated that criticality had no influence and that they approached designing AM parts like any other product. However, most (60%) of respondents noted that AM significantly influenced testing and inspection requirements, including 50% of those who initially stated that criticality did not affect their design approach. The results also indicated that respondents focused on aeronautical applications required design approach adaptations more frequently than those in the space industry. Respondents working across both industries consistently indicated that criticality impacts both the design approach and testing requirements.

The higher the criticality, the more conservative the design approach, with professionals opting for well-known materials and more mature AM processes (R8, 9, 15). Respondent 9 commented on how the design approach would change to ensure the incorporation of multiple load paths for fail-safes. This design approach involved integrating higher safety margins, redundancy and safer geometries to minimise risks due to component design. These considerations influenced decisions such as material selection (R3, 4, 10, 11), wall thickness (R10) and build orientation (R5, 10, 17) to enhance the reliability of AM critical components.

Criticality also introduced challenges, including high testing costs and the time-intensive nature of verification. Stricter testing regimes required methods such as CT scanning (R11, 14, 17). However, Respondent 17 cautioned against excessive testing requirements, stating from their experience that "CT kills the business case" for an AM product due to its high expense.

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Considerable data was collected on the impact of component criticality. However, to maintain the focus of this paper, findings on its influence on the design approach and AM implementation have been briefly summarised here and will be expanded upon in future work. This section has therefore provided a succinct but sufficient answer to the fourth research question.

5. Aerospace AM design approach

Section 4.2 has highlighted that professionals require new design support and knowledge to balance the duality of design benefits and process constraints in AM product development. Given the several gaps identified in existing design support, engineers in the industry were found to adapt their approach to design, moving away from traditional approaches during AM aerospace product development.

The generic PDP by Ulrich *et al.* (2020) describes distinct and sequential stages. This approach works well with traditional manufacturing, where the cost of design changes increases significantly as the process progresses (Cantero-Chinchilla *et al.* 2024). However, implementing AM enables a shift from this approach as the close design-production integration of AM processes blurs the boundaries between the traditional phases. The interview study revealed that respondents described their approach between these phases as highly iterative, with frequent feedback loops enabling designers to adapt and refine their concepts more flexibly than with other manufacturing methods. The findings from the respondents indicated that while the designers follow a broadly similar process when using AM, the design-production integration necessitates an altered approach, incorporating AM-specific design aspects.

To investigate this altered approach, the interview transcripts of the respondents were thoroughly read by the researchers and coded according to the coding structure shown Table A1 in the Appendix using the NVivo software. Once all 20 interview transcripts were coded, all excerpts assigned to code ID 2.4, *Design approach*, were transferred from NVivo into an Excel workbook. The design approach excerpts were then reviewed in detail to identify specific activities of the design approach discussed by the respondents. This analysis led to the iterative development of distinct stages for approaching AM design, drawing inspiration from the models of product development of Pahl & Beitz (2007) and Ulrich *et al.* (2020) and the AM product development model of Dordlofva & Törlind (2018). The excerpts were then grouped into the distinct stages they were evidencing, and through further examination of the sequence in which the respondents described conducting these stages, an order of the stages was recognised. Once arranged, the stages were further refined. For example, *DfAM Principles Integration* was distinguished as a separate stage from *Initial Concept Development*. This division was made because the results highlighted that integrating DfAM principles required considering multiple factors worthy of note outside the scope of early concept development. Figure 5 presents a model of the general aerospace AM design approach consisting of the following steps: 1) Requirements Definition, 2) AM Suitability Evaluation, 3) Initial Concept Development, 4) DfAM Principles Integration, 5) Post-Processing Planning, 6) Build Simulation and AM Design Verification and 7) Prototyping and Testing.

The ordering presented in Figure 5 was constructed based on the frequency with which respondents explicitly mentioned activities occurring in a given order.

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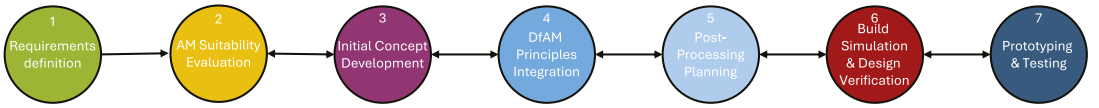


Figure 5. A general model of the aerospace AM design approach.

In this way, the sequence reflects the empirical findings from the interviewees' practices in the context of DfAM in the aerospace industry. For example, several respondents emphasised that material and process selection should be considered during the earliest stages, as requirements are being defined. This process order reflects the close connection between product requirements, such as space-qualified material options, cost targets and qualification constraints, and the feasibility of manufacturing with AM. Materials and process selection in the requirements stage differ from conventional PDP models, such as that of Ulrich *et al.* (2020), where the choice of materials and the definition of production processes typically occur in the Detail Design phase, beginning with some consideration given during earlier stages.

This general model provides an overview of the key activities and considerations of the DfAM process, offering a top-level staged approach synthesised from the interview study findings. This model complements previous research into AM's influence on established product development models, such as Dordlofva (2018), who refines the PDP model to integrate AM design and part qualification through the practice of systems engineering and concurrent manufacturing processes. Additionally, this research complements other studies mapping how the introduction of AM affects the traditional design process, such as the work of Gradl *et al.* (2022) and (Dash, Nordin & Johansson 2024).

Gradl *et al.* (2022) provide an overview of the major process steps in the iterative lifecycle of AM aerospace components through literature reviews, internal NASA studies and academic and industry research. The research in this paper builds on their work by incorporating insights from discussions with AM aerospace industry professionals, offering a representation of the state of practice.

Dash *et al.* (2024) conducted an extensive systematic literature review of DfAM literature that accounts for both the opportunities and constraints of AM. Their research led to the development of a Dual DfAM research landscape model, linking the support literature to the engineering design process. Their model provides a directory of literature on design support for different phases of the design process, depending on the designer's objective, such as conceptual design or parameter optimisation. Their work highlights a scarcity of literature addressing Dual DfAM implementation at an overall process level, particularly in planning and early design phases. The research in this paper addresses this gap by presenting and modelling design and manufacturing considerations throughout the PDP.

The following section presents the analysis and synthesis of the respondents' approaches to designing and developing AM aerospace components. It outlines the sequence and flow of considerations and supports used throughout the design process, from the initial decision to use AM to the final design or redesign of a product. Additionally, it highlights respondents' insights on how customer specifications impact design freedom, the design supports employed and compares design approach stages for new versus redesigned AM products.

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5.1. Requirements definition

Respondents identified defining requirements as the first step in designing or redesigning an aerospace component for AM. As stated by Respondent 20, "it starts from what the product needs to do," as the product requirements define the manufacturing process requirements, ultimately determining the AM suitability and feasibility. Similarly, Respondent 19 stated, "The functional and performance requirements are driving the overall design." Respondent 13 reinforced this, stating, "the most important thing is requirements" because *they must be revisited throughout the design process. This iterative approach reflects the inherent feedback loops in AM design, where the design is refined at each stage based on new insights.*

5.1.1. Material and process selection

In this initial stage, the AM design approach decisions focus on selecting the material for the product and the platform. Platform selection involves choosing the type of AM process and the specific AM machine that can produce the product, with early consideration for the size and orientation of the product. Respondent 11 stated that at this stage, an engineer considers the flow-down of system requirements and their specific component requirements and, based on these, conducts a detailed trade-off analysis of material selection and processes capable of forming those materials to evaluate AM process options.

Material selection was identified as a key initial consideration in the AM design process by respondents 1, 3, 4, 11 and 16. Respondent 1 remarked that material selection is "the first and foremost thing" because "every material in AM performs differently with certain designs." This point was reinforced by Respondent 4, who stated, "some materials are harder to print in than others, so they require further thought." Consequently, material selection is a topic that requires extensive discussion when defining the requirements for AM aerospace product development. Understanding how a material will behave in the AM process is essential for making appropriate design choices. This point was highlighted by Respondent 13, who emphasised that AM is "a very multidisciplinary technology." They further explained that an AM designer must possess more than just AM design knowledge to succeed, stating, "A designer should not just be a designer. They should also have a fundamental understanding of the materials, the inspection, and all that type of good stuff."

Material selection goes together with platform selection during the requirements definition stage. Respondent 4 described these two factors as "big hitters" that require early consideration, as they shape many of the discussions between the designer, product owners and manufacturing engineers. Respondent 11, who works for a governmental space agency, highlighted that their organisation has noted that their organisation has invested in "education on proper process selection and proper material selection" due to frequently encountering poor uses of AM, where inappropriate materials or processes were chosen because they were readily available. According to Respondent 11, it is "a very bad design philosophy" to base decisions solely on the materials and machines available within a company. Instead, designers should "select the [AM process] that makes sense for the application," with due consideration for the material and the product size. As "size determines platform," as stated by Respondent 3.

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Process selection is a critical early consideration for AM products, with several factors to consider. Gradl *et al.* (2022) identified 12 attributes for aerospace component AM process selection that designers should consider. Gradl *et al.* (2022) noted that design requirements may limit the available process options, if not entirely dictate the selection. This limitation is particularly relevant when considering product size, material properties and process maturity requirements. To support designers in understanding the aerospace AM process selection process, Gradl *et al.* (2022) provide a graphic illustrating various AM aerospace processes and examples of their production capabilities. The graphic emphasises the trade-off between build/deposition rate and feature resolution of AM components.

5.1.2. Iterative requirement refinement

Respondents from different organisations highlighted material and process selection as critical early steps in the AM design approach. However, the way each organisation integrates AM-specific considerations into its design process was found to differ. Some organisations adopt a distinct approach for AM, emphasising process-specific requirements early. However, others integrate AM considerations into their existing design approach. Respondent 11 commented that they do not approach additive design differently from other manufacturing processes. Respondent 12 similarly stated that their company has design practices for developing a turbine rather than dedicated design practices specifically for AM. These responses highlight the variability in the AM design approach. Although AM has unique considerations, organisations may maintain a design approach across all manufacturing methods, integrating AM considerations into that approach instead of treating it distinctly. For example, Respondent 12 elaborated that their design approach is "mainly as [in] ordinary product development, but now we have a part with the design for AM as well," explaining that AM-specific considerations may come later in their approach through defining requirements for different levels of testing due to selecting AM.

Once the requirements are set for the product, Respondent 13 explained that they then engage with the product owner and ask, "Did you mean these requirements, or did you actually mean this?" This next stage is where Respondent 13 states, "is what industry calls design for additive manufacturing," as it marks the commencement of developing the product into a design optimised for AM, rather than merely a product produced using AM.

5.2. AM suitability evaluation

Once the requirements of the product have been defined, the designer then evaluates that product's suitability for being produced using AM. Evaluating suitability is where the designer considers the specific design opportunities AM offers compared to traditional manufacturing methods for their product. The activity here is comparable to Design for Manufacturing (DfM), where DfM is defined as the "design of product and process specification for cost-effective, reliable manufacture to achieve customer satisfaction and business success" (Miles 1989). Respondent 13 highlighted this perspective: "In my view, it's just design for manufacturing. The additive part just happens to be the tool."

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Continuing that, at this stage, a designer has "an aspirational design with a fit, form, function, interfaces [and] loads, etc."

This stage requires extensive engagement with the customer/product owner (R1, R2, R10, R13), collaborating to define how the aspirational design can be achieved through the selected AM process to meet the requirements. Hence, Respondent 13 described this as "a partnership development process." Evaluating AM suitability involves investigating where AM can impact the product design or redesign, from both a functional and a product development perspective. For a product being redesigned for AM, Respondent 1 stated that the requirements are taken as a set of design boundary conditions. At this stage, the AM designer reviews the product and considers how a component could benefit through AM opportunities like part consolidation (R7, 13) or improved functionality (R5, R7, R9, R11). Respondent 9 described this as looking at the product "through the lens of benefits," reviewing the design and development for potential benefits such as lead time reduction, increased agility and performance or functional improvements using AM compared to conventional manufacturing. Respondent 9 highlighted that AM's agility makes it particularly suitable for engine components, as it allows fast iterations and design tests without the need to wait for long-lead-time items requiring tooling. They provided specific examples, such as "engine cooling geometry" and "lightweighting."

However, considerations for design feasibility are required to balance the perceived benefits with the product requirements. Using the example of rocket engines, where cooling presents a significant challenge, Respondent 20 described how cooling channel integration drove their design approach when using AM. They noted that the design process started by determining the number and size of the cooling channels (setting the requirements), which ultimately constrained the design space.

This consideration highlights the importance of evaluating suitability, as designers must ensure that leveraging AM design freedom does not compromise product requirements. They need to assess whether the process can achieve the desired design while accounting for limitations such as overhangs, surface roughness and material property variation.

Designers assess the viability of AM for their product in this stage, as Respondent 11 stated, "I also like to say, don't use additive unless you have to," highlighting that "additive has to buy its way in" by demonstrating clear performance improvements or economic advantages, whether in cost or time. Often, they noted, it can provide both. Respondent 11 commented that their governmental organisation is "developing a lot of new materials that you can only produce with additive to have performance advantage." Demonstrating how material selection informs the suitability of a product for AM. Respondent 11 raised other considerations, such as machine limitations and post-processing requirements. Stressing that a designer should "not focus on the build process itself," instead, "design every step of the AM process early on or you will not be successful. You will fail parts."

Respondent 13 also emphasised that product suitability for AM must be considered in relation to the verification and validation routes, as "your means of compliance should be pretty much locked in" when evaluating suitability. Additionally, Respondent 17 highlighted staffing requirements, noting that they "needed 25 people for two parts," signifying that the AM supplier must consider their own feasibility in delivering the product with their resources. These factors

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highlight the importance of early technical and organisational planning and considerations of the impact of AM on the entire PDP from the outset.

Designers need to consider the balance between the complexity of the PDP and the benefits provided to the product using AM. In balancing these considerations and their own AM capability, Respondent 10 stated that it is the AM supplier/engineer's job to then help the customer/product owner/OEM "withstand the risks" associated with implementing AM and to discuss different directions, processes or machine modifications needed.

At this stage, designers are constrained by the limitations imposed by OEMs and the degree of design freedom they have. Respondent 5 highlighted the challenge when a product is originally designed for manufacturing using a different technology, and the OEM does not permit design modifications. Since the AM supplier does not own the design, Respondent 5 states that they cannot modify it to be fully suitable for AM, as any "potential solution using additive, should be with exactly the same geometry," leading to the loss of AM's potential design benefits. In such situations, the suitability assessment focuses more on product development opportunities from a cost, rate, or delivery standpoint. Respondent 5 described this design approach as "modify for additive manufacturing," due to the limited design freedom available to the AM supplier. Hence, the focus is on adapting production strategies rather than redesigning the part for AM.

5.3. Initial concept development

Once the part has been deemed suitable for AM, the next step in the design approach is to develop the initial design concepts. This stage of the design approach varies depending on whether the designer is creating a new product specifically designed for AM or redesigning an existing product to be produced using AM.

5.3.1. Verification and validation considerations

The initial conceptual designs are created and iterated to meet the requirements defined during the first stage. Designers focus on ensuring the design satisfies mechanical and thermal requirements and assessing whether it can meet the margins for these requirements (R11, 19). Hence, during this stage, designers must also consider how to verify and validate that the requirements will be met (R11). AM design verification is the confirmation that the specified requirements have been fulfilled, while validation is defined as confirming that the requirements or a specific intended use or application have been met, both requiring objective evidence (America Makes 2023). According to the ECSS standards, *verification* for space components is the process of demonstrating that a product has been designed and produced according to its specifications and is free of defects. While *validation* is the process of demonstrating that the product can accomplish its intended use in the intended operational environment (ECSS 2023).

At this stage, designers need to consider these aspects and evaluate how their design can achieve the benefits identified during the AM suitability evaluation (R10). With the requirements defining the value of AM from a product end-use perspective, designers in this stage focus on identifying the manufacturing opportunities provided by different concepts (R10). Respondent 10 noted that the initial concept phase can feel "all-encompassing" due to the numerous AM-specific considerations involved compared to traditional aerospace manufacturing

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processes. Stemming from the design-production integration of AM, as a designer tries to fully optimise their part for their process while ensuring it can be validated. Thus, the concept phase in the AM PDP may take longer than in traditional aerospace product development. However, the AM PDP allows for this additional time, as the design-production integration eliminates the time constraints associated with designing and manufacturing tooling once a concept is defined (R10).

5.3.2. New vs redesigned

This distinction between AM and traditional development also influences how new and redesigned products are approached in the PDP. Respondent 15 provided an example involving the early stages of designing concepts for custom AM space brackets. They begin by gathering requirements, boundary conditions and loads, then use algorithmic design in a CAE tool to create the initial concept geometry. Algorithmic approaches, such as topology optimisation, were found to be often employed as a starting point for structural components. Similarly, Respondent 16 explained that for structural components, they begin with their defined design space and collaborate with partners to perform topology optimisation. Once the topology-optimised design is generated, they then review it using aerospace-qualified FEA tools to develop the initial concept.

Similarly, Respondent 16 stated that their design philosophy for redesigned products is to create a design space that includes the original geometric volume and a test case before running a topology optimisation. This approach helps to avoid situations where they are "trying to optimise too many stuff all at once" and risk encountering a local minimum. Respondent 16 described breaking down mechanical environments into separate design and verification environments for the spacecraft actuator AM design. For example, in the initial mechanical design phase, they exclude vibration loads, and instead, the design focuses on proof loads, such as proof pressure for structural components or the ultimate load for electromechanical components. Then, they calculate other test cases in the verification design environment and make design adjustments as needed. This approach enables more targeted and efficient optimisation and is primarily based on the engineer's practical experience.

For redesigned products, some respondents focused on keeping the product geometry as close to the original as possible while identifying areas where modifications could benefit the AM process. This review and modification process was often done in collaboration with the product owner through workshops to determine feasible changes. Respondent 3 suggested starting with the notional design, often a cast design, for redesigned products transitioning to AM production. Some of their customers favoured this approach as they were already changing the manufacturing method and, as noted by Respondent 3, "don't want to add in two variables" in changing the design and the manufacturing process. As a result, their aerospace customers for redesign tended to take a conservative approach to their initial concept design during their implementation of AM, so only the manufacturing process needed recertification. Respondent 3 referred to this approach as adapting the product for AM, that the products are "redesigned for printability … not actually designed for AM." This perspective was supported by Respondent 4, who described it as "design for printability," and Respondent 5, who similarly referred to such designs as being "adapted to AM." In this context, the

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initial concept design focuses on adapting the product to make it printable. For example, Respondent 4 described a scenario where AM was being used to replace a casting process, and the product required redesign because "some features easily castable are not easily additively manufactured." Respondent 4 added that this approach requires continuous engagement with the product owner, asking questions such as, "If we were to make that change, does the part still retain its function?" Within this approach, designers have the freedom to modify elements like support structures to enable the manufacturing of the current design. A discussion with the product owner is then required to ensure that the designed support structures do not interfere with critical surfaces.

When the designer has greater design freedom to redesign a product, the extent to which a designer can modify the product can be constrained by the level of detail available in the original component design supplied by the product owner. This lack of detail can create challenges in determining how much adaptation is possible. Respondent 3 noted that they typically begin with drawings of the parts, which were often prepared for a casting manufacturing process. Consequently, designers "don't necessarily get a great appreciation for what the actual endpoint is," meaning they lack clarity on the final intended product being redesigned for AM. For example, if the casted product requires post-processing to drill holes, a designer may miss the opportunity to propose an AM-adapted design with integrated holes if this detail is not included in the original design. To address these challenges, Respondent 4 stressed the importance of asking design space-defining questions during workshops, such as identifying the product's interfaces. Similarly, Respondent 2 emphasised the value of conducting workshops with key individuals designing the part before running simulation software. This collaborative approach enables design engineers to better understand where modifications can be made to simplify manufacturing in AM. It also ensures that manufacturers clearly understand the design's constraints and what is feasible.

5.3.3. Balancing design expertise and hybrid manufacturing considerations

The quality of the initial concepts is heavily dependent on the knowledge and skills of the designer or engineer. At this stage, a review of the designs by multiple team members is necessary to ensure the optimal use of AM and confirm the feasibility of the concepts. However, a lack of AM knowledge among engineers can present challenges. Respondent 10 highlighted this issue by describing situations where a customer presents them with an AM aerospace part, and their immediate thoughts were, "This is a s**t design. What kind of idiot did this?" only for the customer to clarify later, "It was our AM expert." This issue can arise when an engineer's AM expertise and design focus too narrowly on mitigating a single AM design consideration or achieving a specific requirement without considering how a design choice impacts other requirements or AM-related constraints.

The interviews corroborated the process illustrated in Figure 1, as respondents evidenced that creating a product design involves a continual feedback loop to earlier stages and the parallel development of the manufacturing process. Designers must ensure their chosen process can reliably produce the product to meet requirements. Respondent 10 emphasised this point: "Even though we talk about design, you mainly talk about process as well… and all of a sudden, just talking about design isn't just talking about design." Respondent 17 viewed process

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selection as a part of the concept design phase due to the setup considerations required to get parts into production. These considerations included paperwork for powder management and the processes for material recycling.

At this stage, a designer should also consider how implementing a hybrid manufacturing process could facilitate the creation of a better product or improve the PDP. As Respondent 10 put it, "Stop trying to solve everything within the powder box," highlighting that a design could, at times, be better achieved or improved in terms of cost, time, or performance by combining AM with traditional manufacturing or post-processing. Respondent 10 elaborated that designers should consider, "What are the bits that work well in the AM process, and what are the bits of design that could be done traditionally as an insert?"

Türk *et al.* (2019) provide an example of a hybrid manufacturing approach that combined the design freedom of AM with carbon fibre composites to produce lightweight, complex structures. Demonstrated on a robotic leg structure, their approach achieved a 55% weight reduction and significant lead time savings. With these considerations for a balanced and hybrid design approach, the next stage focuses on refining the design to fully integrate feasible DfAM principles and maximise AM's advantages.

5.4. DfAM principles integration

This next stage focuses on designing the product to maximise the benefits of AM by fully integrating its complexities and adhering to the DfAM principles specific to the chosen process. Design principles are fundamental rules, derived from extensive experience and evidence, that provide design guidance towards reaching a successful solution (Fu, Yang & Wood 2016). Valjak & Lindwall (2021) state that the main purpose of DfAM principles is to support the early design and the realisation of a product design in a form suitable for AM. In this stage, the initial design concept is refined to ensure buildability, fully leverage AM's design capabilities, meet the product's requirements and maximise AM's benefits within the PDP. In this stage, the extended time available during the concept phase, without the usual constraints of tooling development, allows designers to avoid a scenario where a product is designed in a way that only AM can produce it, yet it is not truly designed for AM (R10).

5.4.1. Leveraging design for X in AM

At this stage, respondents were found to follow or apply different Design for X (DfX) techniques to take advantage of AM's interconnected design and production. DfX describes a range of product development techniques that can be applied to achieve concurrent improvements in a product's cost, quality and cycle time (Huang & Mak 1997). Well-established DfX tools like Design for Assembly (DfA), such as Boothroyd-Dewhurst Design for Assembly (Boothroyd 1994) and Lucas Design for Assembly (Miles 1989), encourage designers to address manufacturing and assembly challenges during the design phase, reducing costs and time later in the PDP. DfAM is a subset of DfX, and the respondents noted how implementing AM made it easier to incorporate other DfX techniques. Enabling them to address their designs' most critical success factors through AM's alternative design freedoms. For example, AM complements DfA through enabling the use of part consolidation design methods, which can greatly reduce the number of parts

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needed to produce a component. Which in turn reduces the associated fasteners, interfaces, assembly operations and time. Such design changes are particularly advantageous for aerospace components, where each interface introduces potential failure points requiring inspection and qualification due to high safety and reliability requirements.

Respondent 1 called part consolidation one of the core elements of DfAM for aerospace, as it enables the removal of redundant weight by eliminating subassemblies. Additionally, Respondent 5 highlighted a key DfX consideration for AM, which they called "design for surface finish." As highlighted in Section 4.1.2, surface roughness is a major consideration for AM products due to its potential impact on part performance, geometrical accuracy and downstream post-processing requirements, which increase time and costs (Obilanade *et al.* 2021). The surface roughness condition is influenced by design and process factors, necessitating early consideration during design stages. A "design for surface finish" approach helps optimise surface roughness through considered design choices for features or orientation, improving part performance, meeting requirements, or minimising negative impacts on the PDP. However, Respondent 5 noted that such constraints "break the statement of design freedom" as the designer's choices for design features or part orientations become limited.

Another approach noted by the respondents was *Design for Functionality* (DfF) (R7, 9, 10, 13). In the context of AM, DfF refers to a design approach in which the designer prioritises the intended function of the part over the constraints imposed by traditional manufacturing methods (Zhong, Ornelaz & Krishnan 2017). Respondent 9 described DfF as a designer starting with the *optimal design* and then working backwards to accommodate constraints. They provided an example of an engine component, explaining that they "stripped [the design] back to first principles of what would the perfect cooling geometry look like for this application," before determining how to achieve that geometry with their available manufacturing processes. In this approach, DfF focuses on designing the optimal cooling geometry for the application by first ensuring the best performance from a CFD and thermal dynamics perspective. Then, the part is refined to incorporate the constraints of the chosen AM process, ensuring manufacturability.

Respondent 10 explained that before implementing AM, their space antenna horn clusters were complex assemblies requiring welding, making them prone to discontinuities, failures and manufacturing errors. Using AM to consolidate the design, the final product achieved a 90% weight reduction compared to the original. This advantage is particularly significant in the space industry, where reducing weight translates to substantial cost savings. In 2020, the average launch cost per kilogram to Low Earth Orbit (LEO) was $10,313 for non-commercial satellites and $4,092 for commercial satellites (Adilov *et al.* 2022).

Respondent 7 noted that many customers realise they can integrate more functions into a single AM part, but this comes with trade-offs. Respondent 10 highlighted the compromise between manufacturing and functionality, explaining that AM, like all manufacturing processes, has rules and that "when you break the rules, you have to pay." That payment is in cost, time or both, later in the PDP. For example, Respondent 10 explained that while AM may result in a slightly higher initialisation cost in the antenna cluster consolidation, improved functionality and weight reduction benefits justify the trade-off. One approach to optimising AM designs that requires trade-off considerations is topology optimisation.

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When implemented in AM, it presents unique challenges that require careful consideration, particularly in aerospace applications.

5.4.2. Topology optimisation and aesthetic constraints

For many traditional subtractive manufacturing processes, designers use topological optimisation to determine the most efficient material distribution geometry, treating the optimisation outputs as a conceptual design guide rather than a final design. Designers take the output and refine it to ensure feasibility for traditional manufacturing. This refinement is necessary due to the complexity of achieving the optimised design using conventional methods (Mirzendehdel, Behandish & Nelaturi 2020). With the design complexities offered by AM, some engineers attempt to print topological designs directly.

The outputted topology-optimised geometry can have imperfections due to factors such as mesh discretisation limitations and inaccurate density methods, leading to an inability to account for manufacturing-induced deviations (Pasini & Guest 2019). Respondent 10 noted that some organisations overly rely on these tools in AM, treating them as "black boxes," inputting requirements and pressing print without refining the DfAM buildability or considering other AM performance impacts. Respondent 8 warned against design support that provides solutions in this manner, stating, "You don't see why that solution was chosen or what could be the pitfalls in choosing this."

Even when a topological design output meets the product requirements, aesthetic expectations in aerospace can present additional challenges. Algorithmic design techniques often produce organic-looking structures. Respondents noted that in aerospace, such structures can be viewed negatively due to preconceived notions about what a structurally sound part should look like and concerns about the difficulty of validating and verifying complex product designs. Respondent 10 provided an example where a chief engineer reviewed a customer-prepared part for printing and remarked, "This part will never fly. It's too ugly!" Therefore, Respondent 10 coined the term "too ugly to fly" to describe designs that, despite effectively integrating AM design principles to meet requirements, are dismissed due to their aesthetic and require redesign. Respondent 10 elaborated that their organisation frequently redesigns their customers' AM products to make them more acceptable for aerospace applications. A process they referred to as "two-and-a-half D optimisation," in which they conduct the full 3D topology optimisation and then simplify the design to look like something that could have been machined, that is, it has smooth, flat surfaces and other aesthetic features that inspire greater structural confidence.

5.4.3. Balancing design optimisation and manufacturing constraints

AM design guidelines are applied to understand design limitations and process capabilities at this stage. These guidelines provide insight into constraints such as maximum buildable overhang angles and other process limits, helping designers navigate the many factors affecting a part's design limitations.

Respondent 19 highlighted the various considerations and factors influencing the quality of an AM overhang, such as the bridge distance, print height and layer thickness. They emphasise that in implementing DfAM principles, one should approach the design from a "first-principles understanding of what can and cannot

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be done with the process." Hence, during this DfAM principles integration stage, a designer becomes most aware of the design limitations for their product success imposed by the material, AM process and the specific printer (R10). An understanding of these limitations can be difficult to fully comprehend; thus, achieving optimal designs often requires iterative cycles of design, printing and refinement, providing valuable learning opportunities for new designers.

This stage of the design approach requires manufacturers to balance design limitations with optimisation while managing downstream impacts. These discussions on balancing typically occur during what Respondent 6 described as a "design for manufacturing review." They elaborated that in this review, designers work with their machinists to establish "a certain number of rules regarding the parameters … [and] regarding the technology," helping designers understand limits such as support structure types and sizes. At this stage, a designer with a strong understanding of AM and access to effective design support can leverage the AM process and its perceived limitations to enhance the product or more easily meet its requirements.

Respondent 11 provided an example of designing microchannels in rocket heat exchangers using laser powder DED processes. By understanding process characteristics, such as build angles and their resultant surface textures, designers can tailor these surfaces to benefit the application. For example, to meet specific requirements, such as achieving a specific pressure drop, designers might not need to prioritise designing to achieve as-built smooth internal surfaces or performing post-processing to ensure smoothness. Instead, as suggested by Respondent 11, by accounting for the expected as-built surface roughness within the channel, a designer can modify the channel design to augment the heat transfer or pressure drop by incorporating the surface roughness as a functional element. This approach allows the design to meet specified requirements without requiring adjustments away from the optimal design geometry or post-processing.

Respondent 10 provided an example of needing to assist customers in redesigning topology optimised parts because the client had not considered the post-processing's impact. A topology-optimised part may be designed to handle a load in one direction, but once printed, it may need to be machined on a surface perpendicular to that load direction, where it lacks strength. Therefore, designers must account for how the post-processing machinery will handle the part during surface finishing when implementing DfAM principles and maximising their optimisation. This observation was supported by Respondent 17, who stated, "Just because something is topologically optimised, it's not additive." As designers integrate DfAM principles and balance design constraints, they must also anticipate the role of post-processing in achieving the final product requirements.

5.5. Post-processing planning

Once the design concept is finalised to meet requirements and integrate AM principles, the next major consideration is post-processing. Although post-processing is described as the fifth stage of this proposed approach, consideration for it will have occurred during earlier stages. These considerations include understanding the level of post-processing required to meet the initial requirements and ensuring, during concept development and DfAM integration, that the design limits the need for post-processing while maximising the benefits of AM.

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These considerations mean engineers face challenges in the concept design phase, attempting to account for all factors that influence the product in a single attempt. Respondent 10 highlighted that AM is not about achieving a perfect design on the first try but about experimentation and problem-solving. Hence, as some design and production challenges cannot be fully resolved in design, engineers may need to adapt their designs and plan for challenges to be addressed downstream through post-processing activities, such as adding offsets or regions for clamping.

Respondent 5 summarised this by stating, "To understand design choices, I always recommend to hear the needs for the post-processing activities." Similarly, Respondent 11 stressed the importance of planning every step of the AM process early, warning that "if I miss the post-processing step in my conceptual design, I'm probably going to scrap my part." Once more emphasising the difficult balance of the many design and production variables in AM due to the close design-production integration.

Post-processing planning includes design considerations for material removal at interfaces, powder removal, base plate removal, support removal and inspection. Respondent 11 highlighted that designing for Inspectability (DfI) is "one of the critical areas in AM," urging designers to consider, "How do I inspect [my part] to make sure that I meet the microstructure and geometry [requirements]?" DfI involves optimising manufacturing parameters and design features to enable ease of inspection. It is another DfX approach that can be effectively applied through AM to adapt designs (Cantero-Chinchilla *et al.* 2024). For example, when using Wire and Arc AM (WAAM), the metal grain structures can cause coherent noise that can hide defects when inspecting using ultrasonic methods (Cai *et al.* 2024). Hence, optimising material selection and process parameters is essential to reduce data interpretation issues based on the chosen inspection method. Additionally, geometrical considerations for inspection are important, particularly for generative design solutions. Thus, AM designs may require adaptation to incorporate gaps in key structural areas for safe access and to enable optimal placement of sensing techniques (Cantero-Chinchilla *et al.* 2024).

AM-specific DfI approaches have been proposed, such as the framework by Mahan *et al.* (2022), who describe a process for DfI of AM components when using pulse-echo ultrasonic inspection. Their framework applies AM design heuristics to create designs with increased inspectability. While their framework was found to improve design inspectability, it comes at the cost of strength and mass, requiring a trade-off of requirements, potentially limiting the scope for designing low-weight, high-strength components, one of the key opportunities of AM in aerospace applications. Respondent 11 referred to these considerations collectively as "design for post-processing."

As AM designs become more complex, ensuring buildability and inspectability requires careful planning. Simulation plays a key role in assisting with post-processing planning and design. These tools enable designers and machinists to discuss and address challenges inherent to AM, such as surface roughness, support structures and base plate removal.

5.6. Build simulation and design verification

When designing a product for AM, a designer must understand and consider the part's function, the manufacturing process and how key design and production

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decisions, such as part orientation and support structure, affect the quality of its features. Respondent 10 highlighted that this all-encompassing decision-making is challenging during the concept design phase, as AM design "takes iterations… sometimes you have to scrap [the design] …. and start in a different direction with lessons learned." To support this decision-making, pre-build simulation and process verification are crucial in ensuring manufacturability and optimising the AM process.

5.6.1. Pre-build simulation and process verification

At this fifth stage of the approach, designers use build preparation simulation tools to conduct these design iterations. Simulation will have been performed in earlier stages, including topological optimisation, CFD, stress analysis and other CAE simulations, to ensure the product functions while taking advantage of AM's design complexities. The simulations at this stage are more focused on preparing for AM production, including modifications to the product's design to ensure buildability, such as detailed design of support structures.

Although many considerations for buildability would have been addressed earlier, this stage verifies those decisions through AM pre-build CAE software, such as *Magics* by Materialise NV (R3, 4, 5, 15, 16, 17). Respondents 19 and 20 explained that during this stage, they conduct specific checks on the part geometry, using simulation tools to identify and highlight errors. Using both AM-specific tools like *Magics* or adapting traditional DfM systems to perform specific analyses. Slicing tools were also employed to examine design areas with uncertainties in more detail. Respondent 4 described this stage as "putting [the part design] through the process to get it onto the machine, as the machine likes it."

The printing process is simulated here to identify potential issues that might impact the build's success. As Respondent 3 stated, build simulations provide the user with "an indicative answer as to whether the build will be successful." Respondent 2 gave examples of some factors they examine during this stage, such as features that have "sharp areas," "stress raisers," "recoater collisions," "deflections," "distortions," or are at "risk of cracking." The primary goal of using these simulations is to identify and address problems that could affect the build's success. However, although these simulation tools help indicate the build success, as noted in 4.2.2 by Respondent 4, they help "identify problem areas, but not the magnitude."

Respondent 3 explained that, for example, a simulation might identify an overhanging feature as challenging to build, but cannot specify whether it is entirely unbuildable by a process or will just lead to reduced geometrical accuracy or poor surface condition. Respondents 2 and 5 pointed out that this stage identifies and mitigates issues like distortion. Respondent 2 highlighted that pre-build simulation results allow designers to include pre-compensation in the design. Pre-compensation involves adjusting part features in the opposite direction of the expected distortion, so the part naturally corrects itself as it cools after printing. Similarly, Respondent 5 described addressing distortion by reinforcing supports or including "ghost parts" in overheating zones. Ghost parts are auxiliary structures or features added to manage thermal and mechanical stresses during the build process (Olleak *et al.* 2024).

At this stage, designers collaborate closely with the OEM of the machine being used. Respondent 17 mentioned how they consult with their LPBF machine

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supplier to better understand the machine's capabilities beyond what is available online or in user guides. They emphasised their need for more access to the OEM to ensure they had the correct material files to ensure accurate simulations and reliable production. Material files are digital representations of the physics-based behaviour of the material during the AM process, which help the pre-build software provide accurate simulations of part production.

5.6.2. Support structure design and build plate optimisation

Several respondents explained that simulations are conducted to explore the baseline support requirements automatically generated by CAE AM tools. These provide a starting point for feasibility assessment, after which engineers refine and optimise bespoke supports using their own design knowledge to minimise complexity and volume. This stage, where support structures are designed, was emphasised as one of the areas in which AM professionals have the greatest freedom to determine how to produce the part. This freedom in support structure design was emphasised as a key area for redesigned parts (R2, 3, 17), particularly when an OEM restricts a designer in how much they can adapt a part for AM. Respondent 4 stated they "create supports to avoid changing the part." Respondent 3 highlighted their "flexibility in terms of adding stock material" because downstream manufacturing processes remove these supports. Designers use a combination of support structure design methods, including pre-build software-generated "block supports" and custom-designed CAD supports tailored to the part and process (R3).

When redesigning products for AM manufacturing, some engineers stated that before looking at the requirements, the first activity they do once they receive the part from the customer is to put it into the build preparation software. Respondent 4 stated, "Nine times out of ten – You probably won't open CAD… you'd just open Magics." This sentiment was echoed by Respondent 3, who stated that when redesigning products, "50% of the work's done," and their initial focus is determining "how you're putting it [in the machine] … [and] how many will fit." The respondents stated they wished to maximise the number of parts-per-build to fully utilise the AM machine's capacity, either to reduce costs by achieving high output or to meet part orders without running multiple builds. Respondent 4 highlighted that orientation is a critical factor in maximising output but must be balanced against its impact on buildability. They explained that while one orientation might allow ten, another might only fit six, but could be the better choice to ensure successful builds.

Build simulation is a crucial process stage, requiring significant time and resources, even with expert knowledge and effective design support. Respondent 4 emphasised this, stating, "We shouldn't underestimate that, because I would say if I look at a part… I'll spend three days getting it ready," adding, "I'll spend a day at least having discussions and talking." Highlighting the efforts involved in building a simulation, even after the part design is confirmed.

5.7. Prototyping and testing

Once the initial part design is completed and the build simulation verifies that the part should be successfully built, prototyping and testing occur to investigate design uncertainties and further verify the product. Respondent 16 commented that once this stage of the design approach is reached, it would ideally be "a part

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that goes out from the additive manufacturing machine, and you use it." However, they explained that in practice, the part enters post-processing steps, as planned in earlier stages, where surface treatment and machining are performed to achieve the final design's intended shape. Before this final print occurs, several test prints will have been conducted concurrently with the other design stages. For example, post-processing tests may have occurred for surface roughness treatment considerations, where test coupons are treated to observe the material loss and resultant surface condition. Respondent 4 explained that prototyping and testing are conducted to optimise the many parameters of the AM process, focusing on "honing on those [optimal] parameters, [as] those incremental changes impact on a micro level."

The knowledge gathered from coupon testing can support build preparation and help designers address AM process issues related to build platform placement (de Oliveira *et al.* 2024). Respondent 5 highlighted the issue of porosity varying depending on the part location within the build platform. To address this, Respondent 5 described their approach to this challenge, stating, "Print the whole bed with a lot of coupons" to assess porosity levels and mechanical performance at different spots in the machine.

This process allows the designer to create a "colour map of performance" for a property across the build plate. Respondent 17 described a similar process, highlighting how this type of testing data allows designers to prepare their builds and designs to account for areas of weak properties on the build platform. Allowing them to "design parts around that weakest sort of link," by carefully considering part orientation. The results of prototyping and testing are used to inform necessary adjustments to the design and process, such as adding more material, altering a parameter or modifying orientation, creating a sequential loop of iterative testing and redesign to refine and verify the final design.

Prototyping and testing are listed as the last stage of the design approach because they require a good comprehension of the envisioned design that is ready for evaluation, as each test and artefact iteration print incurs a cost in both time and resources. While the earlier stages utilise design supports to address AM-specific challenges, designers can more comprehensively verify whether the product will meet the requirements and specifications through prototyping and test coupons or artefacts. This concept is eloquently summarised by Respondent 10, who described knowing the moment they successfully addressed the challenges of AM design as occurring "When I print the prototype."

5.7.1. Design and process verification and qualification

The printing of prototypes helps provide confidence in the buildability of a design. However, Respondent 14 added that when prototyping, they often utilise in-line monitoring and inspection tools to gain more confidence in the part itself. Respondent 14 highlighted that underlying defects can go undetected, explaining that while a print may appear successful, "if there is nothing in the machine that tells us there is a lack of fusion, we can't guess it." Hence, there is a need for tools for verifying that the process has operated as planned.

Respondents 6 and 14 explained that they adhere to the ECSS material and process verification standards, citing standard procedures such as tensile and powder testing, including metallography and flowability. Respondent 6 described using a full engine thrust chamber prototype for water flow tests to verify that the

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pressure drops as expected, ensuring the surface roughness impact of the part aligns with the design. They further elaborated that the ultimate test of a part occurs during its use in the rocket. However, numerous qualification tests are conducted before an AM engine undergoes a hot fire test to ensure compliance with ECSS standards. Respondent 6 emphasises that even if the design is validated before the hot fire test, because it is an AM part, "It's always a prototype."

In addition to standard test coupons and ahead of full product printed prototypes, respondents 7, 14 and 16 highlighted the use of product-specific design artefacts that represent sections or features of their design. By isolating a part's design feature into a representative design artefact, one can investigate specific material property issues, such as surface roughness at that feature. Respondent 14 explained that these artefacts act as "a prototype representative of the final shape of the hardware," allowing them to inspect, test and validate the design. Describing a preference for a trial-and-error process or, as Respondent 16 terms, "learning by doing" through these artefacts, as they provide a better representation of the actual design than standard test coupons or simulation tools. As Respondent 7 conveys, "you can't have a representation from like a tensile bar or a tube about the properties because it changes so much in the part."

The use of product-specific artefacts to understand and assess uncertainties in AM design has been explored in the work of Dordlofva & Törlind (2020). Through a research collaboration with space industry companies, they studied three AM product development case studies and proposed the AM design artefact (AMDA) process, further developed into the AMDA method by Obilanade *et al.* (2024). This method describes how AMDAs can be used to investigate design, material properties and manufacturing uncertainties. Dordlofva & Törlind (2020) and Obilanade *et al.* (2024) provide detailed practical space industry-related examples of how these product-specific design artefacts help resolve AM-related uncertainties. Furthermore, although Respondent 7 identified this artefact-based approach as their preferred method for prototyping and testing, they also emphasised that it requires a "big cost of investment" in time and resources.

5.8. Summary of the aerospace AM design approach

The previous sections have outlined a design approach for AM metal components developed from the insights gathered in the interview study with aerospace industry professionals. As seen in Figure 5, the proposed approach is not entirely sequential. After the requirements are set, the results of each stage can be fed back into the previous stage, incorporating new part design or process knowledge that may impact the part's ability to meet requirements or highlight the machine's capability to build the design. Similarities to the generic PDP are noted when comparing the proposed AM design approach's stages with the Ulrich *et al.* (2020) phases. However, the close integration of design and production in AM and the relative ease of late-stage design variation result in a blurring of design phase boundaries compared to those defined by Ulrich *et al.* (2020). Figure 6 compares the activities in the proposed design approach model with the design and manufacturing activities within the phases of the generic PDP by Ulrich *et al.* (2020).

The planning phase corresponds to the proposed approach's activities in stages 1 and 2. These stages involve understanding existing design and manufacturing constraints and their significance. Constraints include the designer's current

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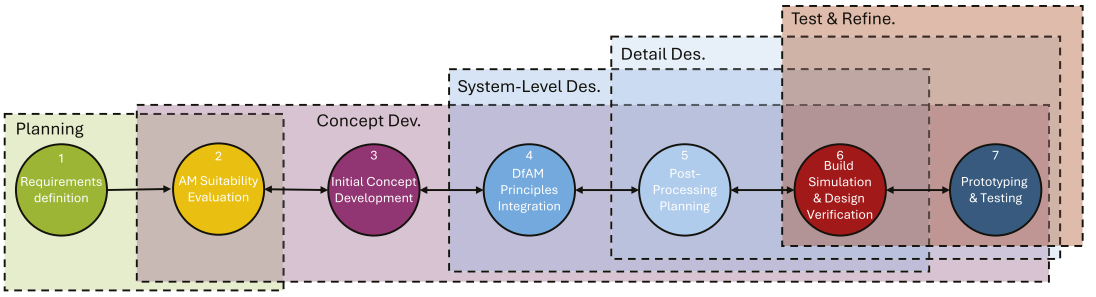


Figure 6. General overview of the AM aerospace product design approach compared with the design process of Ulrich *et al.* (2020).

knowledge of the process's capabilities and economic and time-related considerations when determining the suitability of AM for manufacturing the product. At this stage, the designer considers their supply chain strategy, assesses the technology and defines the product requirements and specifications, including the necessary testing procedures for qualification.

The activities in stages 2–7 of the proposed approach share several similarities with the design and manufacturing activities described in the concept development phase outlined by Ulrich *et al.* (2020). The interview study results evidenced the *all-encompassing* nature of this phase in AM. Several respondents expressed that the activities within stages 2–7 of the proposed AM approach were conducted as part of concept development. The sequence of these activities was often described as iterative, with continuous feedback loops. This iterative nature was particularly evident in the use of build simulation tools and design verification, which were found to be ongoing processes rather than confined to a single stage.

In the Ulrich *et al.* (2020) process, the system-level design phase closely aligns with stages 4 to 6 of the AM design approach. System-level design activities involve defining the product architecture and decomposing the product into subsystems. With AM, specific design approaches and techniques, such as topology optimisation and part consolidation, support this decomposition while simultaneously refining the part's design. From a manufacturing perspective, design considerations for cost occur at this stage, balancing optimising a design for a specific function with the whole AM product production cost.

The detailed design phase corresponds to activities in stages 5–7, where the engineer focuses on ensuring they are designing an AM product, rather than designing a part manufactured using AM. At these stages, designers concentrate on maximising the benefits of their chosen AM process and realising identified opportunities while incorporating process-specific constraints, particularly those related to post-processing requirements. To support this, specification-driven prototyping and testing are conducted to reduce or eliminate uncertainties related to both design and process. While such activities are most prominent in the detailed design phase, initial prototyping and testing typically begin in the concept development phase, where engineers create prototypes to better understand their concepts and the AM processes. Dordlofva & Törlind (2020) emphasise that design artefacts can serve as structured means to identify, explore and reduce AM-related uncertainties in these early stages, thereby supporting more informed design decisions. Pahl & Beitz (2007) provide a model for the design process similar to

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Ulrich *et al.* (2020), noting that correcting fundamental shortcomings in traditional product design at the detail design phase can be extremely difficult or impossible. However, due to the overlapping and dynamic AM design approach enabled by AM's integrated design and production, respondents indicated that such corrections were easier to make in AM.

In the detailed design phase of the Ulrich *et al.* (2020) design process, three key issues are addressed: material selection, production cost and robust performance. Respondents identified material selection as one of the first decisions in the AM PDP, integrating it into the requirements definition, as it significantly impacts AM design feasibility. The use of build simulation software assists designers in estimating production costs based on build time, material usage and post-processing requirements. Finally, prototyping and testing help assess design robustness through design artefacts, coupon investigations and qualification testing. These findings are fed into the detailed design phase to ensure product performance under uncontrollable variations. In the Ulrich *et al.* (2020) process, tooling is developed in the detailed design phase. However, with AM, tooling is not often required. Instead, detailed design considerations for support structures are incorporated at this stage to improve the quality of complex AM builds. Support structure design is a key area of research and development, with studies focusing on developing guidelines for designing and optimising support structures (Vaidya & Anand 2016; Jiang, Xu & Stringer 2018; Langelaar 2018; Paggi *et al.* 2019).

Testing and refinement activities occur iteratively through stages 6 and 7, ensuring that the part design functions as intended and is manufacturable using the AM process. In the Ulrich *et al.* (2020) model, this phase involves two types of prototyping: Alpha prototypes, which assess whether the product will function as designed and meet requirements (often not made using the final manufacturing process), and Beta prototypes, which evaluate performance and reliability for final adjustments (Ulrich *et al.* 2020). With AM, prototyping can occur much earlier, allowing for more rapid and incremental changes. Using prototypes early for quick analysis and refining the prototype design as more knowledge is gained about a problem, allowing for early failure and, thus, quicker success. This approach aligns with the "*rapid, rough, and right*" principles for prototyping described by Lawrence (2003). Dordlofva & Törlind (2020) observed a similar approach when they proposed the AMDA process. They noted that in early AM design phases, engineers used design artefacts to drive the product design specification by building on their insights. Then, as the project progresses, the design of artefacts is specification-driven, as they are used to validate the design specifications.

Notably, the model developed from the interview findings does not include a comparative stage extending into production ramp-up, as seen in Ulrich *et al.* (2020). This stage lies outside the scope of the design approach described by the respondents. Instead, the model captures the process up to the point where a designer is satisfied with the concept and proceeds to qualification and certification through component and system testing. In this sense, the model represents part of the product development and design process rather than the whole process, as system-level testing still follows ahead of ramping up production. Respondent 10 highlighted a key difference between AM and traditional design approaches at the final prototyping and testing stage. In AM, a design can progress through the PDP to production, only for an unforeseen flaw to emerge, potentially leading to a failed part and a scrapped design. Respondent 10 noted the difficulty engineers face

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in accepting this due to the perceived waste of time. However, they pointed out that this is less problematic in AM, as the absence of tooling and the flexibility to modify designs without extensive rework provide greater adaptability.

The proposed approach in Figure 5 offers a top-level overview of the AM design stages identified in this interview study. It provides a detailed description of how AM influences the design approach during product development, effectively addressing the third research question of this study. After developing the general approach model, references from respondents describing alternative approaches were also noted. These were incorporated into a more detailed model of the AM design process. Figure 7 provides an overview of the approaches reported by the respondents, including the iterative cycles and variations in the order of activities described.



Figure 7. Detailed overview of the AM product design approach in aerospace applications.

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The refined model is represented as a circular diagram, illustrating a more iterative design process with multiple variations in possible paths through the design stages. As shown in Figure 7, progression through the modelled stages was not linear for all respondents. Some sidestepped certain stages or engaged in iterative loops between earlier and later stages before continuing with the modelled approach. The arrows between stages in Figure 7 indicate which respondents described transitions between those stages in that way, with the thickness of the arrows representing the number of respondents who stated that direction in their design approach.

For example, a commonly identified iterative loop was 4–6–5–4, where engineers verified the integration of DfAM principles with the buildability of their part and incorporated this feedback into their post-processing planning. This loop was particularly evident when assessing the limits of unsupported overhangs while balancing the effects of surface roughness.

Despite these variations, the overall design approach followed by most aerospace professionals aligned with the direction presented in Figure 5.

6. Discussion

The following sections present a discussion and overview of the themes addressed in this paper and the study's research questions, connecting the findings to wider literature and practice in AM, product development and aerospace.

6.1. Opportunities and applications of AM in aerospace

To address the question, "What are the current applications of AM in aerospace, and what opportunities and challenges do they present for product development and design?" this study highlights several insights. AM offers opportunities by enabling the design of complex, lightweight geometries that improve aerospace product efficiency through part consolidation, enhanced functionality and cost reduction. Specific application opportunities include integrated cooling channels, improved thermal performance and optimised rocket nozzle design. AM also supports faster design and production iterations, helping to mitigate long lead times and supply chain challenges common in the aerospace sector (Dordlofva, Lindwall & Törlind 2016). However, while AM increases agility in product development, its adoption must be balanced against cost and business value. In some cases, AM is viewed less as an immediate, cost-effective solution and more as a long-term strategic investment, particularly as future capabilities evolve and to strengthen the company's image or positioning.

Additionally, many of the challenges identified in this study align with those reported in AM design support literature. These include the difficulty in achieving a precise understanding of material properties in AM parts (Yadollahi & Shamsaei 2017), the challenges associated with qualification and the lack of established standards (Lee, Nagalingam & Yeo 2021) and issues related to support structures and surface roughness (Calignano & Minetola 2019; Artzt *et al.* 2020). Similarly, recurring issues included product distortion and the challenges of managing material and process variability (Haapalainen *et al.* 2020). These results emphasise the need for considerations in balancing the design opportunities afforded by AM with the constraints imposed by the manufacturing processes. For instance, while

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AM enables the creation of intricate, part-consolidated geometries, designers must temper their designs to account for post-processing requirements and support removal considerations (Zink *et al.* 2020). This balance between design capability and feasibility necessitates an integrated design approach, where engineers consider downstream implications during the early design phase, highlighting the importance of DfAM support.

6.2. Aerospace-specific AM barriers

Despite the study being conducted within the aerospace industry, it was noted that many of the design and product development challenges described by respondents were AM-specific, not industry-specific. The main challenges affecting the development of AM components for aerospace applications are equally relevant to designers and engineers in other sectors working with AM processes for high-performance products. The primary aerospace-specific challenges from the study were the standards, regulations and qualification requirements that aerospace engineers must adhere to and the role of regulatory agencies in providing guidance and support. Similar challenges would likely apply to other industries with high-performance critical parts, such as medicine, which must also comply with its respective regulatory and industrial standards when manufacturing components using AM.

However, due to the required level of confidence in an aerospace component, this lack of regulatory understanding, coupled with the design challenges of AM, affects its adoption in aerospace, particularly for critical applications, due to the costly current methods for attaining that confidence, such as CT scanning. Among respondents, there was a notable split in perspectives; 60% stated that component criticality influenced their design approach, while 40% indicated that it did not. Those who reported that criticality influenced their design approach highlighted that it led to more conservative design choices, prioritising well-known materials and additional safety features. Specific standards have been developed for critical components to address these challenges and provide guidance on the operation and production control of AM machines and processes for aerospace applications. An example is ISO/ASTM52904-19 (2019), which helps establish best practices for ensuring reliability in critical LPBF aerospace components. These results help address the research question, "how does component criticality affect design considerations for AM?"

6.3. Overcoming design and regulatory challenges

The respondents identified several different supports and approaches to overcoming the design challenges of AM in aerospace. For the regulation challenge, many respondents discussed how collaborating with agencies like ASTM has been one way to adapt, such as Respondent 7, who worked with their customers to "adapt ECSS standards for AM constraints." Work has been conducted on identifying the standards needed to help adopt AM (Lee *et al.* 2021). However, there are still gaps in standards provisions for design challenges such as surface roughness (Obilanade 2023).

Respondents emphasised using collaborative approaches to overcome AM-related issues in product development. Product design review workshops

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with cross-disciplinary teams were highlighted as a particularly effective means of identifying and addressing manufacturing issues early in the design process. As highlighted by Wiberg *et al.* (2019), who conducted a comprehensive review of design supports for AM and stressed the importance of involving various experts early in the design process to address design issues to ensure the feasibility and performance of a product align. As found by Wiberg *et al.* (2019) and this study, one of the most common approaches to overcoming AM challenges is using CAE tools to understand design capabilities and limits, model components and simulate the process. These design support tools were found to help the designers attain a clearer visualisation of their design workflows and provide feedback on the outcomes of design decisions. Further, tools such as topology optimisation software were highly valued in the aerospace industry for aiding respondents in using AM's capabilities. These tools are well documented for their ability to help designers meet structural integrity and performance requirements while reducing material needs and helping produce low-weight components (Hurtado-Pérez *et al.* 2023).

Despite the availability of advanced methods and tools, significant limitations and gaps in design support were identified. A lack of comprehensive material property data within design support systems restricts engineers' ability to make informed decisions. This limitation impacts the accuracy of simulation methods, reducing the confidence in the results acquired, particularly in meeting the stringent certification needs required for aerospace applications. As a result, the continued use of prototypes remains essential to gain specific knowledge that cannot currently be attained through computational means due to the gaps in material data.

6.4. Towards an integrated AM design approach

Through an analysis of the respondent's approaches to designing AM aerospace products, the paper has presented a general model of the AM design approach that considers both design and manufacturing aspects in product development. Each stage of the proposed approach could be further expanded with more detail and direction. For example, in the requirements clarification stage, one could be directed to apply a framework for AM process selection for aerospace components, such as that described by Gradl *et al.* (2022), or a decision-making methodology for AM material selection in aerospace applications, as outlined by Junaid *et al.* (2024). One could also be directed to use the MPDS of Hajali (2024) for AM suitability evaluation. The ISO/ASTM 52910 (2018) standard provides an overall strategy model for DfAM, focused on criteria such as cost, quality and delivery time, along with a procedure for identifying AM potential. In contrast, the models developed in this paper embed suitability evaluation directly within the requirements definition stage, extending the design approach beyond the standard's initial feasibility check. Additionally, various DfX approaches can be prescribed, such as the DfI method proposed by Mahan *et al.* (2022) while applying DfAM principles.

Further, the study identified loops in the design approach that were implemented by respondents when deviating from the general sequential approach, for example, looping from the integration of DFAM principles to prototyping and testing and back to update the post-processing plan to optimise the feasibility of the design. A formalisation of this design loop has been presented by Obilanade *et al.* (2024),

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who provide a detailed method to investigate design issues through design artefacts as prototypes. A detailed formalisation of other loops within the design approach, such as an AM suitability assessment to a design verification loop, would be useful in helping engineers understand the value added by AM for their product and in making a go-or-no-go decision for its use.

An important aspect not explored in this study is the use of AM from a sustainability perspective. Many aerospace companies advertise AM as a sustainability-driven development approach (Orbex 2022; Villamil *et al.* 2018). Hallstedt *et al.* (2023) illustrate how companies can leverage AM for sustainable product development in aero-engine manufacturing, reducing fuel consumption and material waste. Reducing fuel consumption and material waste is beneficial in aerospace from both cost and environmental. Additionally, social sustainability concerns arise from the procurement of critical materials from politically unstable regions (Hallstedt *et al.* 2023). Hence, AM considerations for sustainability could also be implemented, requiring additional support and further altering the design and manufacturing approach.

Sustainability was rarely mentioned by respondents, despite its prominence in public discourse on AM for aerospace. Only two respondents commented on sustainability: Respondent 8 highlighted the potential for reducing a product's footprint by using less material, while Respondent 17 noted that AM can enable more sustainable mould production by replacing carbon fibre composites with faster, lower-cost alternatives. That sustainability was not more widely raised is notable, as it suggests a difference between public discourse and current aerospace engineering practice. This absence may partly reflect the interview guide, which did not include an explicit question on sustainability, as well as the prevailing focus in aerospace practice on cost, performance and qualification over environmental goals. Future research should therefore explore in greater depth how sustainability considerations are, or could be, integrated into AM product development in the aerospace industry.

7. Conclusions and future work

The purpose of this paper has been to explore the state of practice in designing and developing AM aerospace components. Through interviews with 20 aerospace industry professionals experienced in AM, the study first identifies key opportunities and challenges in DfAM for aerospace applications. The identified product opportunities, summarised in Table 2, demonstrate where AM can enable improved product performance, enhanced supply chain efficiency and cost reductions in aerospace applications. Conversely, the identified challenges, summarised in Table 3, reveal technical, operational and organisational barriers to AM adoption in aerospace product development, limiting its broader implementation. These challenges include ensuring the feasibility of optimised AM product design, which remains a key obstacle to fully leveraging AM's potential. These confirm the understanding of aerospace opportunities such as topology optimisation, part consolidation and weight reduction (Gradl *et al.* 2018; Blakey-Milner *et al.* 2021; Khorasani *et al.* 2022) and established limitations such as surface roughness and qualification/certification challenges (Blakey-Milner *et al.* 2021; Kerstens *et al.* 2021; Zhou *et al.* 2021).

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This paper has described how aerospace professionals are supported in achieving these opportunities and addressing challenges. Two types of design supports were identified:

- Practical design supports, summarised in Table 4
- Computational design supports, summarised in Table 5.

The paper expands current AM understanding by providing a detailed description of both practical and computational design supports, while also identifying areas where support is lacking. These results directly address the second research question, "What types of design support are available to industry professionals for AM aerospace components, what determines their effectiveness, and what gaps exist in current design supports."

Despite the availability of simulation software and mathematical topology optimisation tools, several deficiencies and limitations were identified, making them sometimes ineffective in supporting AM design. Hence, traditional design workshops and rapid prototyping remain widely used throughout the AM PDP. By examining the deficiencies in the capabilities of the design supports, gaps for future design support development have been presented in Table 6. These include the need for:

- A design collaboration community.
- A single-suite CAE for AM.
- Surface roughness simulation.
- AI-driven generative design tools.
- Improved file transfer quality between computational supports.

The identified design opportunities, process challenges and design support are not industry-specific. Thus, the findings may be generalisable to other AM-driven industries, such as medical devices and automotive manufacturing. For instance, surface roughness and DfAM principles are critical considerations for qualification challenges and component performance in the medical device industry. Likewise, in automotive manufacturing, AM's benefits in lightweight structure design and tooling reduction are applicable and subject to trade-offs involving cost, material constraints and manufacturability.

However, despite the generalisability of certain aspects of this research, its aerospace-focused context limits direct applicability to industries with different priorities. Additionally, while the respondents were knowledgeable in AM and held various roles within their companies, the study was limited as much of the data was collected, and discussions on AM challenges and opportunities focused on using LPBF. Future research should, therefore, focus on other industries and specific AM processes to enable cross-sector comparisons and assess the transferability of these insights to validate the relevance of the proposed design approach across sectors.

Despite 20 respondents being a strong sample for qualitative work, additional participants would have increased the scope and representativeness of the study. The findings were also limited by the relatively small number of OEM or system-level design respondents, which could have provided further insights. Hence, the component-level focus of this study may limit the generalisation of findings to system integration or whole-system product development. As an interview study, the data is based on engineers' accounts and may reflect self-report bias, with some

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respondents emphasising frustrations or challenges and others highlighting successes. Complementing interviews with observations or case studies could help address this limitation in future work.

In addition to the practical applications, this research contributes to developing knowledge in AM design support and product development theory. This study adds to the research on AM's impact on product development by presenting how AM provides flexibility in design through the ease of rapid design iterations and alternative prototyping approaches, reshaping traditional paths to a successful PDP.

The paper has highlighted how AM requires distinct frameworks and processes for design to take full advantage of the benefits it can bring. These frameworks must include considerations that enable balancing design functionality, qualification requirements and customer considerations with AM's alternative time and economic costs. For example, time and costs may be saved due to the elimination of the need to lock down a design and create tooling. However, new time and cost considerations arise in understanding the balance between surface roughness, performance and post-processing requirements and capabilities, ultimately altering the design approach. The description of the modelled approach provides new knowledge on pathways for DfAM during product development. From the perspective of aerospace, this highlights the need to adapt established product development models, such as Ulrich *et al.* (2020), when implementing AM. This study, therefore, provides an answer to the research question, "How does AM influence the overall design approach during product development?", specifically in the context of aerospace applications. While these findings may not directly apply to all industries, they offer insights that could inform the adaptation of PDPs in other sectors where AM is being introduced, grounded in the perspectives of experienced practitioners.

While the expert opinions of the respondents provide valuable insights, they may also introduce subjectivity due to the respondents' backgrounds and environments, such as their roles as tier-one suppliers, consultants, or software developers. Complementary studies would also be worthwhile to investigate potential biases arising from specific backgrounds. These investigations could include surveys targeting software developers, design case studies comparing the perspectives of OEMs and AM aerospace manufacturing companies, or industry surveys and experiments to provide more quantitative data, enhancing the robustness and applicability of the findings.

Further, the research indicated that component criticality influenced the design approach in some cases. Further investigation into the impact of component criticality on the AM design process is needed to enable transferability for other high-performance component industries. This research will allow engineers to better understand how a component's criticality may influence design decisions and outcomes.

Due to the rapidly changing and developing nature of both the aerospace industry and AM technologies, some challenges may be easier to overcome with the development of new design supports. However, new challenges may also arise, stressing the need for periodic investigations into AM implementation to enable the dissemination of current best practices in AM design and further AM's adoption in the aerospace industry.

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Appendix

The following list provides examples of the interview questions that were asked during the semi-structured interviews:

Respondent Background

1. Could you tell me a little about yourself? (Background, length of employment, position at the company) Years of AM experience
    a. *[Respondent is shown* Figure 2*]* – Here is a model of the PDP for AM. Could you please look at this picture and cross where you would say you work?
    b. And at what tier would you say your company operates?

Q2 AM Aerospace Design

2. What type of products does your company design for AM?
    a. Are these new or redesigned products?
    b. How do you approach designing these products?
3. Why was AM selected for these products?
4. What are the challenges when designing AM Aerospace products?
5. When designing, how do you know that you have overcome these challenges?
    a. Do you test/qualify/certify your AM components?
6. To what extent is a product's design controlled by customer specifications vs your design freedom?
7. Do you use the notion of critical components?
    a. How does your organisation assess the importance, or criticality, of a component?
    b. How does the criticality of a component influence your design process?

Q3 Design Support

8. What resources, tools or methodologies do you/your team use during the AM component design process? (process simulations, etc.)
9. What do you consider is a good design support, what are you looking for in good design support?
10. How do these resources and tools for AM design differ from those used for traditional design within your company?
11. Are there any challenges or limitations with using the existing design supports?
12. How do you think these challenges could be addressed or the design supports improved?
13. How do you assess the impact of the design support? (on the quality of the product or the ease of the design process?)

Q4 Closing

14. Is there anything that you think I've missed?
15. Do you have any questions for me?
16. Can I come back to you if another question pops up?
17. Finally, is there anyone you recommend I interview that you think could have helpful input?

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Table A1. Coding memo used in the analysis

	Code name	Definition
1	Respondent Background	
1.1	Background	What has the respondent done career and education wise
1.2	Length	How long have they worked in their role or AM
1.3	Position	Their current title in their company
1.4	Product Development Stage	Descriptions of the product development stages that the respondent has worked at.
1.5	Company Tier	Any description of how the company operates
1.6	Customer	Name or description of the customers of AM products
2	AM Aerospace Design	
2.1	Product Types	Type of aerospace product the respondent has worked with
2.2	New Products	Specifically related to the design of a new product
2.3	Redesigned Products	Specifically related to the design of a redesigned product
2.4	Design Approach	Related to the approach to designing AM products
2.5	Reason for AM Selection	Why AM was selected for the manufacturing of a product
2.6	Design Challenges	Design challenges for AM engineers
2.7	Overcoming Challenges	How a challenge is overcome
2.8	Testing/Qualification/Certification	Description of methods of testing, qualification or certification
2.9	Customer Specifications	Any specifications highlighted by customers
2.10	Design Freedom	Any discussion on the freedom or restriction to design choice
2.12	Critical Components	Any discussions regarding critical components
2.13	Component Importance Assessment	Method or discussion of how to assess the criticality of components
2.13	Influence of Criticality on Design	How the criticality of a component influences the way the engineer designs or brings about specific considerations for product development
3	Design Support	
3.1	Design Support/Tools/Resources Used	Any resource mentioned that is used for designing AM aerospace components
3.2	Good Design Support Characteristics	Any description of what makes the design support useful
3.2	Differences from Traditional Design	Any comment on the difference between traditional design and AM design supports
3.2	Challenges/Limitations of Design Supports	Any comment on the challenges/gaps/limitations of the design supports
3.2	Suggested Improvements	Any comment on how design support can be improved or enhanced
3.2	Impact Assessment of Design Supports	Any comment on how an assessment or comparison of a design support is done
4	Interesting Quote	Any interesting quotes that could be used within the final document

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